NOTICE
OF CANCELLATION

MIL-C-38240/14A(USAF)
NOTICE 1
25 March 1988

MILITARY SPECIFICATION

COMPUTER, ALTITUDE, ALTITUDE
ENCODING CPU-66/A-14

MIL-C-38240/14A(USAF), dated 12 June 1987, is hereby canceled without replacement.

Preparing activity:
AIR FORCE -71

Project No:
6610-F226

AMSC N/A

FSC 6610

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NOTICE
OF CANCELLATION

MIL-C-38240/22A(USAF)
NOTICE 1
3 April 1988

MILITARY SPECIFICATION

COMPUTER, ALTITUDE, ALTITUDE ENCODING
CPU-66, GENERAL SPECIFICATION

MIL-C-38240/22(USAF), dated 21 December 1970, is hereby canceled without replacement.

Preparing activity:
AIR FORCE -71

Project number:
6610-F227

AMSC N/A

FSC 6610

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NOTICE
OF CANCELLATION

MIL-C-38240/25A(USAF)
NOTICE 1
1 April 1988

MILITARY SPECIFICATION

COMPUTER, ALTITUDE, ALTITUDE
ENCODING CPU-66/A-25, A-4F

MIL-C-38240/25A(USAF), dated 8 March 1968, is hereby canceled without replacement.

Preparing activity:
AIR FORCE -71

Project No:
6610-F229

AMSC N/A FSC 6610

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MIL-C-38286A (USAF)
15 December 1977
SUPERSEDING
MIL-C-38286(USAF)
2 July 1964

MILITARY SPECIFICATION

COMPUTER, FLIGHT DIRECTOR CPU-65/A

This specification is approved for use by Code 99, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for one type of universal flight director computer, designated CPU-65/A.

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Box, Wood, Cleated-Plywood
PPP-B-636	Box, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-D-1000	Drawings, Engineering and Associated Lists
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: (Hq AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016) by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6610

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MIL-C-5541	Chemical Films for Aluminum and Aluminum Alloys
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series, General Specification for
MIL-R-8323	Receiver, Radio R-540/ARN-14C
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-G-25597	Gyroscope, Displacement, Roll and Pitch, Type MD-1
MIL-C-26485	Control Assembly, Attitude Gyroscopic AF/A24G-1
MIL-H-26689	Horizontal Situation Indicator AQU-2/A (Indicator, Multiple, Air Navigation AQU-2/A)
MIL-I-27193	Indicator, Attitude ARU-2B/A
MIL-C-27205	Control Assembly, Attitude Gyroscopic A/A24G-5
MIL-R-27226	Receiving Set, Radio AN/ARN-67
MIL-I-27619	Indicator, Attitude ARU-11/A
MIL-H-27848	Horizontal Situation Indicator AQU-4/A

STANDARDS

MILITARY

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods for Aerospace and Ground Equipment
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals

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DRAWINGS

AIR FORCE

64F 1866 Block Diagram - Computer CPU-65/A.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 PREPRODUCTION. This specification makes provisions for preproduction testing.

3.2 SELECTION OF SPECIFICATIONS AND STANDARDS. Specifications and standards for necessary commodities and services not specified herein shall be in accordance with MIL-STD-143.

3.3 MATERIALS

3.3.1 FUNGUS-PROOF MATERIALS. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

3.3.2 NONMAGNETIC MATERIALS. Nonmagnetic materials shall be used for all parts except where magnetic materials are essential.

3.3.3 TOXIC AND CORROSIVE FUMES. The materials, as installed in the computer and under the service conditions specified herein, shall not liberate deleterious or corrosive fumes. This shall include any fungicidal agents that are used.

3.3.4 NONFERROUS MATERIALS. Nonferrous materials shall be used for all parts except where ferrous materials are essential.

3.3.4.1 Nonferrous materials contained within hermetically sealed inclosures shall be considered suitably protected against corrosion. Requirements specified for fungicidal and corrosion protective treatment and anodizing of aluminum alloy

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parts shall not be applicable for parts within hermetically sealed inclosures. Steel parts within such inclosures shall be cadmium plated in accordance with QQ-P-416, type I, class C.

3.3.5 METALS. Metals shall be corrosion-resistant or suitably treated to resist corrosion from fuel, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.6 DISSIMILAR METALS. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.7 PROTECTIVE TREATMENT. When materials are used in the construction of the computer that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.4 DESIGN AND CONSTRUCTION. The computer shall be designed to provide computed flight director signals to flight director indicators.

3.4.1 The computer shall be so designed and constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.2 RELIABILITY. The computer shall have a mean-time-between failures (MTBF) of 600 hours when tested in accordance with MIL-STD-781B, Test Plan II criteria.

3.4.3 MAINTAINABILITY. The computer shall be so designed that overhaul, including replacement of worn parts, adjustment, balance, and calibration may be accomplished by instrument and electronic repairmen provided with simplified tools and equipment, trained in the repair of electronic and similar equipment.

3.4.4 HERMETIC SEALING. The method of hermetic sealing and mounting of the components in the sealed chamber of the case shall be such that the case can be unsealed and the complete assembly of components removed as a unit, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures unless they are approved by the procuring

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activity. The case and seal (see 6.3.1) shall be designed to meet the pressure requirements specified herein.

3.4.4.1 FILLING MEDIUM. The filling medium shall be at least 90 percent pure, free of dust particles, and shall contain no more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The filling medium shall be either 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The absolute pressure of the filling medium in the case shall be approximately 1 atmosphere. Where practicable, the 100 percent helium filling medium shall be utilized.

3.5 PERFORMANCE. The computer shall be capable of satisfactory operation when subjected to the following conditions:

a. Temperatures - operating temperatures ranging from -54° to +71°C and storage temperatures ranging from -65° to +98°C.

b. Humidity - relative humidity up to 100 percent

c. Altitude - pressures ranging from 30 inches Hg down to 0.315 inch Hg (approximately 100,000 feet)

d. Salt Fog - exposure to simulated salt sea atmosphere for 48 hours

e. Vibration - vibration with an applied double amplitude of 0.060 inch through the frequency range of 10 to 55 cps and 10g input from 55 to 500 cps

f. Fungus - fungus growth as encountered in tropical climate

g. Sand and Dust - sand and dust as encountered in desert areas

h. Rain - rainfall as encountered in any locale

i. Temperature Shock - alternate immersions in water maintained at 85° ± 4°C and 5° ± 4°C for a total of 8 cycles without leakage

j. Acceleration - acceleration forces of 20g without damage

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k. Extreme voltage and frequency variation - applied voltages of 105V at 320 cps and 125V at 480 cps.

l. External pressure - external pressure of 26.5 psia with no power applied

3.5.1 ELECTROMAGNETIC INTERFERENCE. The computer shall comply with the requirements of MIL-STD-461A, Notice 3, for Class A1 equipment for test methods CE03, CE04, CS01, CS02, CS06, RE02, RS02, and RS03.

3.5.2 DIELECTRIC STRENGTH. There shall be no insulation breakdown when 500V dc is applied between isolated pins and between pins and the case for a period of 10 seconds.

3.5.3 LEAKAGE. The initial maximum leak rate at a pressure differential of approximately 1 atmosphere shall not permit a loss of more than 10 percent of the total filling medium after 1,000 hours.

3.6 PART NUMBERING OF INTERCHANGEABLE PARTS. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.7 ELECTRONIC PARTS. Electronic parts and application thereof shall be in accordance with MIL-E-5400. Parts that do not appear on approved lists shall not be used unless approved by the procuring activity. Electronic tubes shall not be used.

3.7.1 OTHER COMPONENTS. All other components in the electronic system shall conform to applicable specifications, where existent.

3.8 MECHANICAL DESIGN. The computer shall be composed of one module for each flight mode listed in 3.10 plus any other components required to provide the operation specified herein. Functionally related modes of simple form may be combined in one module if approved by the procuring activity. Modern techniques of miniaturization and modular construction shall be exploited to the greatest extent possible without sacrificing ruggedness, reliability, and service life. Transistorized components shall be used throughout the computer.

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3.9 CASE. The computer case shall be in accordance with figure 1 and shall be hermetically sealed except for external adjustments which shall be gasket sealed under a dust cover.

3.9.1 ADJUSTMENT PANEL. An adjustment panel shall be located on the front of the computer case as shown on figure 1. The potentiometer adjustments specified in table I and Drawing 64F1866 shall be located on this panel. Each potentiometer shall be properly numbered in accordance with table I. The test points shown on Drawing 64F1866 shall also be located on the adjustment panel to permit isolation of failures within the computer. The test points shall also provide for shorting out of various circuits in the computer for test purposes. Prior to fabrication, the design of the adjustment panel shall be approved by the procuring activity.

3.9.1.1 TEST POINTS. The test points shall be as follows:

a. TJ1 - Shorts out direct roll when connected to TJ6 to allow for measurement of roll rates.

b. TJ2 - Shorts out radio displacement when connected to TJ6 to allow for measurement of radio rates.

c. TJ3 - When shorted to TJ6, vertical pointer flag comes into view.

d. TJ4 - Shorts out course-zero and rates when connected to TJ6 to permit measurement of course gain. Also, shorts out roll rates for measurement of roll displacement.

e. TJ5 - Shorts out pitch-zero when connected to TJ6 to allow for measurement of direct pitch gain.

f. TJ6 - Signal Ground.

g. TJ7 - When shorted to TJ3, vertical pointer flag comes into view (reserved for systems where multiple logic is available for testing vertical pointer warning flag (circuit) - Manufacturers option)).

h. TJ8 - Spare.

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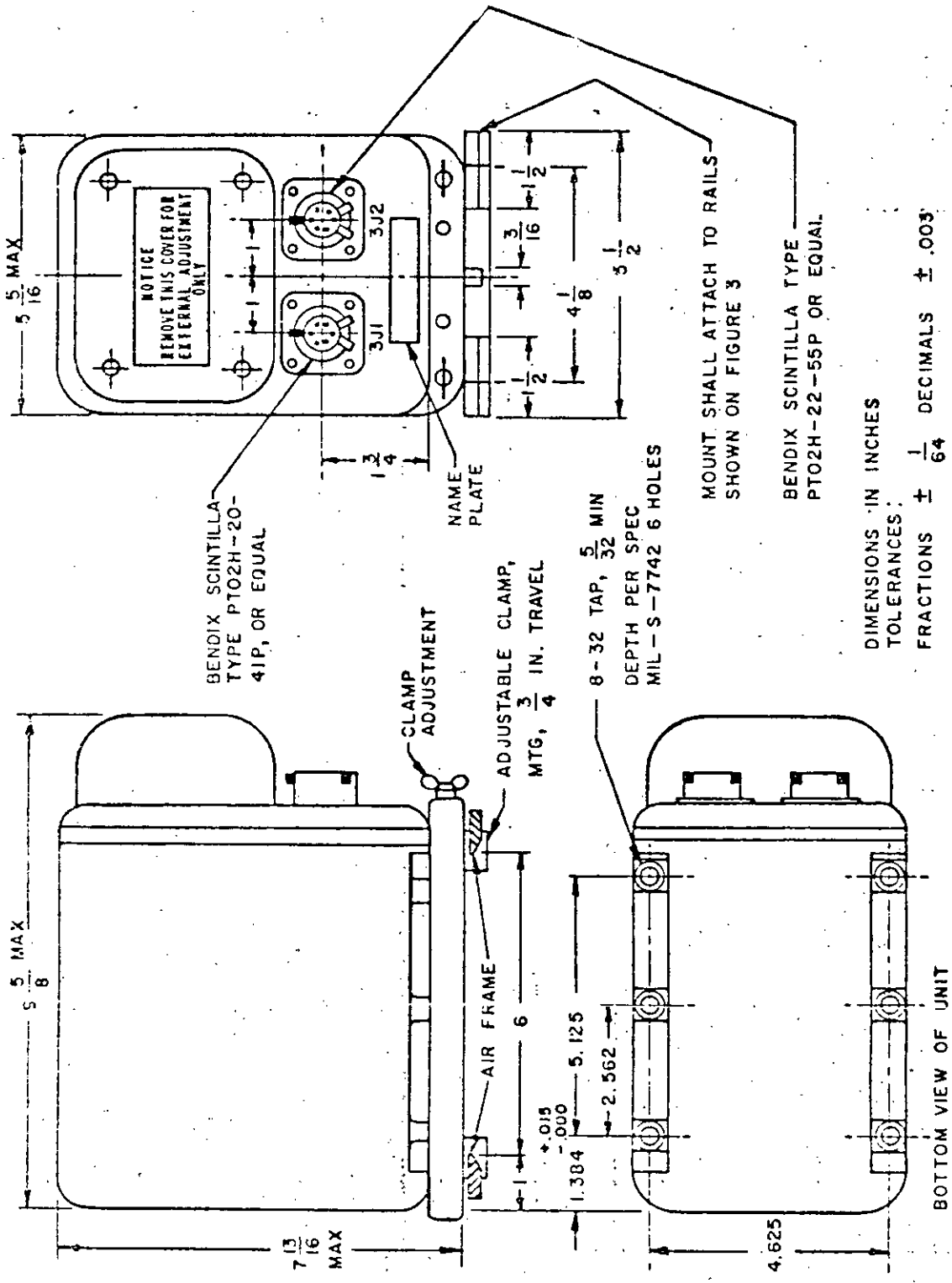


FIGURE 1. Computer, Flight Director CPU-65/A

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TABLE I (See Notes)

POTENTIOMETERS LOCATED ON FRONT PANEL

POT NO.	CONTROL	MODE	MINIMUM RANGE	NOM SET	TOL AT 20° C	TEMP TOL	REMARKS	TOLERANCES FOR VOLTAGE AND FREQUENCY VARIATIONS			
								480 cps 125V	320 cps 105V	420 cps 125V	380 cps 105V
R_1	Approach V.P. Centering Trim	Approach	±0.25 ma	0	±0.02 ma	±0.06 ma		±0.03 ma	±0.03 ma	±0.03 ma	±0.03 ma
R_2	LP Pitch Zero	Approach	±0.06 ma	0	±0.03 ma		Pitch Input 3° (0.6V)	±0.12 ma	±0.09 ma	±0.09 ma	±0.075 ma
R_3	Altitude Error Gain	Altitude Hold	0.24 to 0.75 ma	0.485 ma	0.06 ma		Altitude Input 250 ft (2.48V)	±0.08 ma	±0.08 ma	±0.07 ma	±0.07 ma
R_4	Radio Track Course Cut	VOR	20° to 65° (7.7 to 20.2V)	45° (15.9V)	±3° (±0.8V)	±4°	Radio Input 1.0V	±5° (1.4V)	±5° (1.4V)	±5° (1.4V)	±5° (1.4V)
R_5	Vertical Centering	Heading	±0.25 ma	0	±0.02 ma			±0.03 ma	±0.03 ma	±0.03 ma	±0.03 ma
R_6	Horizontal Centering	Approach Altitude Hold	±0.25 ma	0	±0.02 ma	±0.06 ma		±0.03 ma	±0.03 ma	±0.03 ma	±0.03 ma
R_7	Heading Error Gain	Heading	0.05 to 1.1	0.28 ma	±0.06 ma	±0.12 ma	Heading Input 6.6 (2.58V)	±0.1 ma	±0.1 ma	±0.08 ma	±0.08 ma
R_8	Course Error Gain	Approach	0.025 to 0.55 ma	0.18 ma	±0.02 ma	±0.04 ma	Course Error Input 5° (1.95V)	±0.05 ma	±0.05 ma	±0.030 ma	±0.030 ma

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TABLE I (CONT)

POTENTIOMETERS LOCATED ON FRONT PANEL

POT NO.	CONTROL	MODE	MINIMUM RANGE	NOM SET	TOL AT 20° C	TEMP TOL	REMARKS	TOLERANCES FOR VOLTAGE AND FREQUENCY VARIATIONS			
								480 cps 125V	320 cps 105V	420 cps 125V	380 cps 105V
R_9	ILS Course Cut Limit	ILS	20° to 50° (7.7 to 17.2V)	45° (15.9V)	±2°	±4°	Loc Input 1.0V	±5° (1.4V)	±5° (1.4V)	±5° (1.4V)	±5° (1.4V)
R_{10}	Radio Rate Gain	ILS	0.125 to 0.375 ma	0.325 ma	±0.03 ma	±0.06 ma	Loc Input 30 mv at 0.0159 cps	±0.06 ma	±0.06 ma	±0.04 ma	±0.04 ma
R_{11}	Bank Rate Gain	ILS	0.016 to 0.48 ma	0.32 ma	±0.05 ma	±0.1 ma	Bank Input 10° (2.0V) at 0.0159 cps	±0.07 ma	±0.07 ma	±0.06 ma	±0.06 ma
R_{12}	Course Rate Gain	ILS	0.125 to 0.375 ma	0.25 ma	±0.03 ma	±0.06 ma	Course Error Input 5° (1.95V) at 0.0159 cps	±0.04 ma	±0.04 ma	±0.03 ma	±0.03 ma
R_{13}	Beam Sensor Trip Level	ILS Approach	+10 to +50 mv and -10 to -50 mv	+15 mv and -15 mv	+1.5 mv -1.5 mv	±3 mv		±5 mv	±5 mv	±5 mv	±5 mv
R_{14}	Radio Track Bank Limit	VOR	20° to 50° (4.0 to 9.05V)	25° (5.0V)	±3° (.6V)		Course Error 90° (22.5V)	±5° (±0.95V)	±5° (±0.95V)	±5° (±0.95V)	±4° (0.8V)

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TABLE I (CONT)

POTENTIOMETERS LOCATED ON FRONT PANEL

POT NO.	CONTROL	MODE	MINIMUM RANGE	NOM SET	TOL AT 20° C	TEMP TOL	REMARKS	TOLERANCES FOR VOLTAGE AND FREQUENCY VARIATIONS			
								480 cps 125V	320 cps 105V	420 cps 125V	380 cps 105V
R_{15}	Altitude Hold Pitch Limit	Altitude Hold	5° to 20° (1.0 to 4.0V)	12° (2.4V)	+2° (±0.4V)		Altitude Input (11.8V)	+3° (0.6V)	+2° (0.4V)	+2.5° (0.5V)	+2° (0.4V)
R_{16}	Approach Pitch Limit	Approach	5° to 20° (1.0 to 4.0V)	10° (2.0V)	+1° (±0.2V)	±1.5°	G.S. Input 1.0V	+1.5° (0.3V)	+1.5° (0.3V)	+1.5° (0.3V)	+1.5° (0.3V)
R_{17}	Data Link Bank Limit	Data Link	30° to 60° (5.9 to 10.2V)	60° (10.2V)	+4° (0.41V)	±5°	Heading Input 90° (22.5V)	+5° (±0.5V)	+5° (0.5V)	+5° (±0.5V)	+5° (±0.5V)
R_{18}	Manual Heading Bank Limit	Manual Heading	20° to 60° (+4 to 10.2V)	25° (5.0V)	+2° (0.4V)	±3°	Heading Input 90° (22.5V)	+3° (0.6V)	+3° (0.6V)	+3° (0.6V)	+3° (0.6V)
R_{19}	ILS Bank Limit	ILS	20° to 40° (4 to 7.6V)	25° (5.0V)	+2° (0.4V)	±3°	Course Error 90° (22.5V)	+5° (0.95V)	+3° (0.6V)	+5° (0.95V)	+3° (0.6V)
R_{20}	ILS APPR Bank Limit	ILS Approach	10° to 30° (2.0 to 5.9V)	15° (3.0V)	+1.0° (0.2V)	±1.5°	Course Error 90° (22.5V)	+2° (0.4V)	+2° (0.4V)	+2° (0.4V)	+2° (0.4V)

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TABLE I (CONT)

POTENTIOMETER LOCATED ON FRONT PANEL

POT NO.	CONTROL	MODE	MINIMUM RANGE	NOM SET	TOL AT 20° C	TEMP TOL	REMARKS	TOLERANCES FOR VOLTAGE AND FREQUENCY VARIATIONS			
								480 cps 125V	320 cps 105V	420 cps 125V	380 cps 105V
R_{21}	Doppler Track Angle Error	Doppler	0.05 to 1.1 ma	0.36 ma	±0.04 ma		Doppler Track Angle Error Input 5° (1.95V)	±0.1 ma	±0.1 ma	±0.1 ma	±0.1 ma
R_{22}	ILS Appr Pitch Trim	ILS Approach	±15°	-3°	±0.5°		Nose Down				

NOTES:
1. Bank, altitude, and pitch voltage analogs based on 11.8V maximum output at 90°(270°).
2. Heading and course error and doppler track angle error voltage analogs based on 22.5V maximum output at 90°(270°).
3. Specifications in "ma" are requirements to drive each load.
4. Test equipment error is not considered in the establishment of these values.

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3.9.2 MOUNTING BASE. The computer shall incorporate a mounting base as shown on figure 1 for quick disconnect from the airframe. The mount shall be affixed to the computer case by means of screws and shall incorporate a thumb-screw adjustment so that it may be rigidly clamped to a mounting bracket. The mounting bracket shall be furnished by the airframe contractor and shall conform to figure 3. The quick-disconnect mount shall be such that it can be removed from the computer case and reversed in direction with respect to the case.

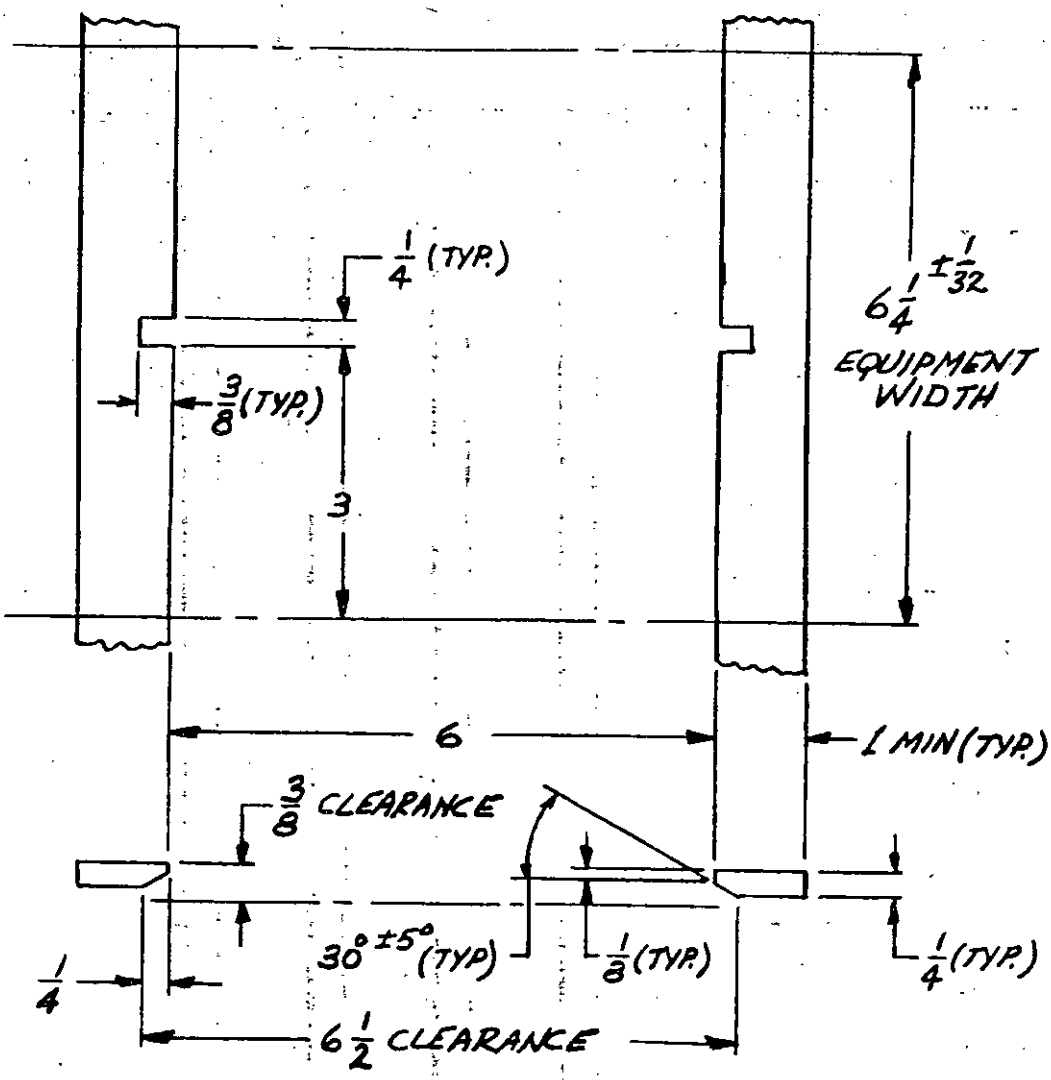
3.9.3 ELECTRICAL CONNECTORS. Two hermetically sealed connectors shall be affixed to the computer as shown on figure 1. The pin connections shall be in accordance with Drawing 64F1866 Connector 3J1 shall be a Bendix PT02H-20-41P, or equal, which mates with a PT06-20-41S, or equal. Connector 3J2 shall be a Bendix PT02H-22-55P, or equal, which mates with a PT06-22-55S, or equal.

3.10 FUNCTIONAL DESIGN. The computer shall supply proper signals to remote flight director indicators such as the ARU-11/A conforming to MIL-I-27619 and ARU-2B/A conforming to MIL-I-27193 attitude director indicators. The computer shall calculate the proper signals for display on the attitude director to enable the pilot to perform proper control action to intercept and maintain the desired course of flight path for the major modes of VOR, TACAN, doppler, data link, and ILS, and in the submodes of MANUAL HEADING and ALTITUDE HOLD. The submodes may be energized during any major mode operation. A NAV mode shall be provided in the computer for biasing all computed information out of view in the attitude director. The computer design shall be such that the final approach mode may be selected either externally by means of a 28V d-c signal or internally by means of a beam sensor when a jumper is provided between pins s and t of plug 3J1.

3.10.1 MODE RELAYS. Mode relays shall be incorporated into the circuitry in such manner as to provide the various modes of operation specified herein when the respective relay is energized by 28V dc and controlled by an external switch. Momentary interruption of the relay circuit shall not result in unsatisfactory signal output when power is reapplied. The beam sensor design shall trigger the final approach mode when the aircraft is on the glide-slope beam and the signal is within $\pm$15 mv of zero.

3.10.2 FLIGHT MODES. The computer shall provide the following modes of operations:

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MOUNTING RAILS TO BE SUPPLIED BY AIRFRAME MANUFACTURE.

DIMENSIONS IN INCHES
UNLESS OTHERWISE SPECIFIED
TOLERANCE: FRACTIONS ±1/64

FIGURE 2 MOUNTING RAILS

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3.10.2.1 BASIC NAV MODE. During this operating mode, no external 28V relay excitation shall feed the computer. The following functions shall be displayed:

a. Vertical steering pointer - out of view to right

b. Horizontal steering pointer - out of view to bottom

c. Glideslope displacement bar - out of view to top

d. Vertical pointer warning flag - out of view

e. Glideslope warning flag - out of view

f. HSI warning flag - out of view.

3.10.2.2 MANUAL HEADING MODE. The flight director shall be placed in this mode when relay K-1 is excited with 28V dc from an external source. The following functions shall be displayed:

a. The vertical steering pointer shall receive a signal of heading error and bank angle. The correct zeroing action shall achieve the desired heading. The amount of required bank angle shall be limited. All sensitivities and limits shall be in accordance with tables I and II.

b. Vertical pointer warning flag - computer signal valid

c. All other pointers and flags shall be out of view as specified in 3.10.2.1 except when placed in view by excitation of a major mode.

3.10.2.2.1 FIRE CONTROL MODE. The flight director shall contain provisions for increasing the sensitivity of the vertical pointer to 7/8-inch deflection for 12° roll. This shall be accomplished by connecting a shunt resistance external to the computer in parallel with the roll ciruitry as shown on Drawing 64F1866. The shunt resistance shall consist of a nominal 3K-ohm fixed resistor in series with a 10K-ohm potentiometer and a relay. When required, the shunt resistor and relay shall be the responsibility of the airframe manufacturer. Both input wires from the shunt shall be shielded and both wires shall be disconnected when not in the fire control mode. The heading error signal shall also be shorted out external from the computer when in this mode. Operation of this mode shall result when the external relay and relay K-1 are energized.

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TABLE II (See Notes)

INTERNAL CONTROLS

CONTROL	MODE	MINIMUM RANGE	NOM SET	TOL AT 20° C	TEMP	REMARKS	TOLERANCES FOR VOLTAGE AND FREQUENCY VARIATIONS			
							480 cps 125V	320 cps 105V	420 cps 125V	380 cps 105V
Pitch Gain	Approach	0.5 to 0.72 ma	0.66 ma	±0.06 ma		Pitch Input 4° (0.8V)	±0.12 ma	±0.09 ma	±0.09 ma	±0.075 ma
Bank Gain	Heading	0.35 to 0.55 ma	0.50 ma	±0.06 ma		Bank Input 20° (4.0V)	±0.09 ma	±0.09 ma	±0.075ma	±0.075 ma
Course Zero	Approach	±0.05 ma	0	±0.04 ma	±0.05 ma	Course Error Input 10° (3.9V)	±0.07 ma	±0.07 ma	±0.07 ma	±0.07 ma
Localizer Radio Gain	ILS	0.3 to 0.37 ma	0.34 ma	±0.04 ma	±0.08 ma	Loc Input 60 mv	±0.06 ma	±0.06 ma	±0.05 ma	±0.05 ma
Glideslope Gain	Approach	0.4 to 0.48 ma	0.44 ma	±0.06 ma		G.S. Input 60 mv	±0.07 ma	±0.07 ma	±0.06 ma	±0.06 ma
Approach Radio Rate Gain	Approach	0.145 to 0.250 ma	0.22 ma	±0.02 ma		Loc Input 15 mv at 0.08 cps				
Approach Bank Rate Gain	Approach	0.2 to 0.24 ma	0.22 ma	±0.02 ma		Bank Input 20° (4.0V) at 0.08 cps				

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TABLE II (CONT)

INTERNAL CONTROLS

CONTROL	MODE	MINIMUM RANGE	NOM SET	TOL AT 20° C	TEMP	REMARKS	TOLERANCES FOR VOLTAGE AND FREQUENCY VARIATIONS			
							480 cps 125V	320 cps 105V	420 cps 125V	380 cps 105V
Approach Course Rate Gain	Approach	0.25 to 0.32 ma	0.28 ma	±0.02 ma		Course Error Input 3° (1.17V) at 0.08 cps				
Time Constant	Approach	8 to 15 sec	13 sec	±2 sec			±2 sec	±2 sec	±2 sec	±2 sec

NOTES: 1. Bank, altitude, and pitch voltage analogs based on 11.8V maximum output at 90°(270°).

2. Heading and course error and doppler track angle error voltage analogs based on 22.5V maximum output at 90°(270°).

3. Specifications in "ma" are requirements to drive each load.

4. Test equipment error is not considered in the establishment of these values.

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3.10.2.3 DATA LINK MODE. The flight director shall be placed in the data link mode when relay K-4 is excited. The pointers shall operate as follows:

a. The vertical steering pointer shall receive a signal of heading error and bank angle as in the manual heading mode. The bank angle limit required shall be as indicated in table I

b. Vertical pointer warning flag - computer signal valid

c. All other pointers and flags out of view as specified in 3.10.2.1.

3.10.2.4 RADIO TRACK MODE. The computer shall provide tracking of VOR, TACAN, or doppler courses. The amount of bank angle required shall be limited. All sensitivities and limits shall be in accordance with tables I, II, and III. A separate mode relay shall be provided for each navigational aid with the following excitation:

a. VOR - 28V dc on pin B of connector 3J2

b. TACAN - 28V dc on pin B of connector 3J2 and 28V dc on pin d of connector 3J2

c. Doppler - 28V dc on pin B of connector 3J2 and 28V dc on pin CC of connector 3J2 if a doppler set without a warning flag signal is used. If a warning flag signal is used, 28V dc on pins B and U of connector 3J2. The following functions shall be displayed in radio track mode:

(1) The vertical steering pointer shall receive a computed signal composed of radio deviation (VOR, TACAN, or doppler) plus course error and bank angle. When the deviation signal is high, the zeroing action called for shall result in a course-cut intercept angle, as specified in table I. The signal shall be such that the pointer will deflect in the direction in which control action must be taken and will center when the aircraft is flying on the desired track or on the programmed flight path to attain the desired track. Crosswind compensation shall be provided so that the track can be maintained.

(2) Vertical pointer warning flag - computer signal is valid

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TABLE III (See Notes)

DEPENDENT GAINS

CONTROL	MODE	MINIMUM RANGE	NOMINAL SETTING	TOL AT 20° C	REMARKS
VOR Radio Rate Gain	VOR	0.15 to 0.43 ma	0.38 ma	±0.15 ma	VOR input 30 mv at 0.0159 cps. Set by the ILS radio rate gain control
TACAN Radio Rate Gain	TACAN	0.15 to 0.43 ma	0.38 ma	±0.15 ma	TACAN input 30 mv at 0.0159 cps. Set by the ILS radio rate gain control
Radio Track Bank Rate Gain	VOR	0.17 to 0.51 ma	0.34 ma	±0.13 ma	Bank input 20° (4.0V) at 0.0159 cps. Set by ILS bank rate gain control
Radio Track Course Rate Gain	VOR	0.27 to 0.81 ma	0.54 ma	±0.06 ma	Course input 20° (7.7V) at 0.0159 cps. Set by ILS course rate gain control
VOR Radio Gain	VOR	0.30 to 0.37 ma	0.34 ma	±0.07 ma	VOR input 60 mv. Set by loc radio gain control
TACAN Radio Gain	TACAN	0.30 to 0.37 ma	0.34 ma	±0.07 ma	TACAN input 60 mv. Set by loc radio gain control
Doppler Radio Gain	DOP	0.22 to 0.27 ma	0.25 ma	±0.03 ma	Doppler input 15 mv. Set by loc radio gain control
VOR and TACAN Course Error Gain	VOR/TAC	0.05 to 0.15 ma	0.10 ma		Course input 10 mv (3.9V). Set by ILS course rate gain control

NOTES: 1. Bank and pitch voltage analogs based on 11.8V maximum output at 90° (270°).

2. Headings and course error and doppler track angle error voltage analogs based on 22.5V maximum output at 90° (270°).

3. Specifications in "ma" are requirements to drive each load

4. Test equipment error is not considered in the establishment of these values.

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(3) HSI warning flag - 28V dc on 3J2-CC-255 $^{+245}_{-0}$ ua (off scale voltage); 28V dc on 3J2-U - connected to AN/ASN-35 computer warning flag

(4) The remaining pointers and flags shall be out of view as specified in 3.10.2.1.

3.10.2.5 ALTITUDE HOLD. This mode shall provide ease in maintaining a desired altitude. The altitude hold mode shall be so designed that it may be selected optionally as a submode during any of the major modes in which the horizontal steering pointer is driven out of view. In the ILS mode, means shall be included for automatic override of altitude hold information at glideslope interception and glideslope beam flying thereafter without manual switching by the pilot. The following functions shall be displayed:

a. The horizontal steering pointer shall receive a signal of altitude error and pitch angle. The correct zeroing action shall cause the selected altitude to be maintained. The sensitivities and limits shall be as specified in tables I and II.

b. All other pointers and flags shall be out of view as specified in 3.10.2.1 except when placed in view by excitation of a major mode.

3.10.2.6 ILS MODE. The computer shall be established in this mode when 28V dc is on pin K of connector 3J1 and when the radio track function is deactivated in the computer as a function of ILS. The following information shall be displayed on each pointer and flag:

a. The vertical pointer steering signal shall mix localizer deviation, course error, and bank angle. When the localizer deviation is high, the zeroing action called for shall result in a course-cut intercept angle as specified in table I. The correct zeroing action shall result in an intercept of the localizer beam. Crosswind compensation shall be provided. Sensitivities and limits shall be as defined in tables I and II

b. Horizontal steering pointer - out of view at bottom

c. Glideslope displacement bar - directly connected to glideslope receiver

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d. Glideslope warning flag - directly connected to glideslope receiver

e. Vertical pointer warning flag - affected by localizer radio signal and computer.

3.10.2.7 ILS APPROACH. The computer shall be established in this mode when three conditions are satisfied, namely, when 28V dc is on pin K of 3J1 to K-2 (ILS) relay, when 28V dc is on pin s of 3J1 to K-3 (ILS approach) relay, and when the radio track function is deactivated in the computer as a function of ILS. Relay K-3 may be excited externally or by the internal beam sensor when jumped between two pins as shown on Drawing 64F1866. The ILS approach mode shall override the altitude hold mode. The functions displayed shall be as follows:

a. The vertical steering pointer shall mix localizer deviation and course error rate, radio rate, and bank rate through the crosswind filter and bank angle as shown on Drawing 64F1866. The bank angle limits and sensitivities shall be as defined in tables I and II

b. The horizontal steering pointer shall mix glideslope devaition and pitch as shown on Drawing 64F1866. Pitch limit and sensitivities shall be as defined in tables I and II

c. All other pointers and flags shall operate as in the ILS mode.

3.10.3 BEAM SENSOR. The computer shall contain an internal beam sensor to automatically switch from the ILS normal to the ILS approach mode when the glidepath is intercepted from above or below the beam. The beam sensor output shall be 28V dc and shall provide proper operation when a wire is jumpered between pins s and t of plug 3J1. A logic circuit shall be incorporated that will require the following conditions to be satisfied before switching. When these conditions are met, the beam sensor output will remain at pin t until either condition a or f is no longer met.

a. Computer in ILS - normal mode

b. Usable glideslope signal (glideslope flag-alarm signal tie-in)

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c. Usable localizer signal (localizer flag-alarm signal)

d. Aircraft within ±2 dots deviation (±150 mv) of the center of the localizer beam

e. Switching shall occur only at a specified deviation from center of glideslope beam (see table I)

f. Radio receiver tuned to localizer frequency (28V dc output from VOR/LOC receiver).

3.10.3.1 HORIZONTAL STEERING POINTER OUT OF VIEW. After switching to the ILS approach mode, the horizontal steering pointer shall be in view or out of view in accordance with the following conditions. Under the horizontal-steering out-of-view conditions, the computer will continue to function in the ILS approach mode.

a. Glideslope and localizer flag signal - The horizontal steering pointer shall be biased out of view with no delay with an invalid flag signal and driven into view with no delay with a valid flag signal. An invalid flag signal is 180 mv or less and a valid flag signal is 255 mv or greater. These values are based on the operation of the ADI and correspond with operation of the flags in the actual aircraft

b. Glideslope deviation signal - The horizontal steering pointer will remain in view with increasing glideslope deviation signal up to 150 mv and will be out of view when the signal exceeds 190 mv. After the horizontal steering pointer goes out of view (due to glideslope deviation), reduction of the glideslope deviation signal to less than the value specified in table I will allow the horizontal steering pointer to return to view.

c. Localizer deviation signal - The horizontal steering pointer will remain in view with increasing localizer deviation signal up to 150 mv and will be out of view when the signal exceeds 190 mv. After the horizontal steering pointer goes out of view (due to localizer deviation), it shall return to view when the localizer deviation is reduced to 150 mv provided the glideslope deviation signal is less than the value specified in table I.

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3.10.3.2 INCOMING RADIO SIGNALS. The incoming radio signals shall be sufficiently filtered within the computer to eliminate carrier noise interference and to provide proper switching when operating with the receivers specified in 3.11.3. Neither false nor nuisance switching shall occur.

3.11 INPUT SIGNALS. Input signals shall have the characteristics specified herein and loading shall be as follows:

3.11.1 ROLL AND PITCH ANGLE SIGNALS. Roll and pitch angle input signals shall be in the form of separate transmitter-synchro 3-wire stator outputs in accordance with the standard test transmitter specified in 3.11.4, with dial readings set at 180° for zero roll and pitch displacement. Increasing dial readings shall correspond to right roll and pitchup maneuvers. Decreasing dial readings shall correspond to opposite maneuvers. The load placed on these signals within the computer shall be equivalent to high impedance, balanced loads. The loads measured between any 2 lines of these inputs shall be at least 5K ohms. The computer shall operate satisfactorily with the MD-1 (MIL-G-25597), AF/A24G-1 (MIL-C-26485), or A/A24G-5 (MIL-C-27205) gyroscope.

3.11.2 HEADING AND COURSE ERROR SIGNALS. Heading and course error signals shall be in the form of separate synchro rotor outputs from the heading datum and course datum control transformers of the AQU-2/A and AQU-4/A horizontal situation indicators in accordance with MIL-H-26689 and MIL-H-27848, or equivalent. The output voltage shall be zero for zero heading error and zero course error. If the actual heading is to the left of the selected heading and course, voltage H prime to C shall be in phase with power excitation ground-to-phase. If the actual heading is to the right of the selected heading and course, voltage C to H prime shall be in phase with power excitation ground-to-phase. Loading of these input signals shall be 10,000 ohms, or greater. The datum control transformers shall be Clifton Precision CTC-8-A-4, or equal.

3.11.3 RADIO SIGNAL INPUTS. Computer loading of radio-deviation, flag, and displacement signals shall be 1,000 ±3 percent ohms each. The computer shall operate satisfactorily with input signals from the following equipment:

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a. Glideslope receiver, Radio Receiving Set AN/ARN-18, or Receiving Set, Radio AN/ARN-67 (MIL-R-27226) and standard ARINC glideslope receivers

b. VOR/localizer receiver, Radio Receiving Set AN/ARN-14 (MIL-R-8323) and standard ARINC VOR receivers

c. Radio Set AN/ARN-21 plus Collins Radio Company 161B-1 TACAN coupler, or equal, Radio Set AN/ARN-65, or OSTER 9616-13 TACAN coupler, or equal

d. Standard ARINC doppler computer.

3.11.4 STANDARD SYNCHRO TEST TRANSMITTER. The standard calibrated synchro, with minimum and known errors, shall be an Eclipse-Pioneer type AY201-1, or equal, high-precision transmitting synchro. A 0° to 360° dial clamped to the rotor of the synchro shall be settable, rotatable, and readable through 360° of rotation to 6 minutes of arc. The following procedure shall be used to calibrate the test synchro:

3.11.4.1 With the rotor leads designated H and C and with one stator lead designated Z and connected to C, 26V shall be applied to H and C (C being grounded). The dial shall be positioned on 0° and clamped to the rotor when:

a. The voltage across H and Y is maximum

b. The voltage across the remaining two stator leads (X and Y) is minimum

c. The voltage across X and Z increases before it decreases for increasing heading indication of the dial. The voltage X to Z and Y to Z shall be in phase with the excitation voltage C to H when the dial reading is zero. For increasing heading indications of the dial, the voltage X to Y shall increase and be in phase with the excitation voltage C to H. The test transmitter shall be set at an index reference to zero and positive rotation reference XYZ.

3.11.5 ALTITUDE ERROR. The altitude error input signal shall be in the form of separate transmitter-synchro, 3-wire stator outputs in accordance with the standard test transmitter specified in 3.11.4. The remote altitude hold sensor shall be synchro driven from the altitude shaft through a clutch so that the

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synchro will remain in the index reference position except when the altitude hold-energize lead is excited with 28V dc from the external mode selector switch. The output shall be 1.0 ±0.1V at 100 feet measured across 5,000 ohms. The range of the synchro shall be 0 to ±500 feet. The computer shall be fully compatible with ARINC standard altitude hold signals provided for flight director and autopilot operation.

3.12 OUTPUT SIGNALS. The computer output signals shall not be affected by those input signals that are not used in the particular mode of operation. To determine the affect of unused input signals not being used in the selected mode shall be set to their maximum value, the input signals used in the selected mode set to zero, and the output shall be measured. There shall be no signal outputs when the above conditions exist. The computer output signals shall have the following characteristics when loaded as specified.

3.12.1 VERTICAL AND HORIZONTAL POINTER SIGNALS. Each vertical and horizontal pointer output signal shall drive two meter movements. The signal shall be direct current with instantaneous polarity and magnitude, as required for the particular mode activated. No a-c ripple voltage of the on-scale signal (as read on a ballantine Model 300A voltmeter, or equivalent) shall be greater than 0.43 times the d-c output voltage or 150 mv, whichever is greater. The off-scale current shall be so filtered that the ripple voltage peak-to-peak shall not exceed 10 percent of the average d-c value. The attitude-director meter-movement loads applied to the output signals shall be driven out of view when supplied with 11.0, -1.0 and +2.5 ma. The resistance of each meter movement shall be 1,000 ±3 percent ohms. A deflection of 7/8 inch from the center position shall require 2.2 ±5 percent ma and shall be the normal full-scale command. The nominal output impedance for the vertical and horizontal steering signals shall be approximately 50 to 200 ohms.

3.12.2 FLAG OUTPUT SIGNALS. Each flag output signal shall be that required to operate 2 meter movement flags having the following characteristics. In those modes in which the flags are driven out of view, the computer shall supply a signal of 255, +245 and .0 ua. In the ILS and ILS approach modes, the glide-slope warning flag shall be connected directly to the glide-slope receiver. In the event of (1) a failure of the B+ voltage

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to an amplifier, (2) an internal short of an amplifier, or (3) inadequacy of the radio deviation signal, the output power normally supplied to pull the vertical pointer warning flag out of view shall cease and allow the flag to appear. The vertical pointer warning flag shall be in view with a radio deviation signal input of 180 ua or less and shall be out of view with a radio signal input of 275 ua or more. This shall be applicable to all modes except the NAV and altitude hold modes. The resistance across each flag alarm circuit shall be two 1,000-ohm loads in parallel ±3 percent. Test points shall be brought out to the front panel to permit adequate testing of the flag circuits incorporated within the computer.

3.12.3 DISPLACEMENT POINTER. The displacement pointer output signal shall drive 2 meter movements requiring 500^{+500}_{-0} ua, each to be driven completely out of view. The sensitivity shall be 75 ua per dot with a total of 2 dots. The load of each meter movement shall be 1,000 ±3 percent ohms. In the ILS mode, the glideslope receiver shall provide the signal directly to the displacement pointer through the computer.

3.13 POWER. The computer shall operate from single-phase, 115V, 400-cps power supplied from an a-c aircraft electrical system in accordance with MIL-STD-704. In addition, the computer shall comply with the requirements of MIL-STD-704 for utilization equipment. The computer shall function satisfactorily with variations in voltage and frequency from 105 to 125V and 320 to 480 cps. An external 28V d-c source shall supply mode relay requirements. The a-c power consumption shall not exceed 16 va at rated voltage and frequency. The power factor shall be not less than 0.85. The d-c power consumption required for relay excitation shall not exceed the following values for each mode at rated voltage:

NAV	0w
Data link	2.3w
Manual heading	2.3w
Radio track	10.0w
Altitude hold	4.5w
ILS	8.5w
ILS approach	13.0w

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3.14 *DIMENSIONS AND TOLERANCES.* Dimensions and tolerances not specified shall be as close as is consistent with best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the computer, they shall be held or limited accordingly.

3.15 WEIGHT. The weight of the computer, with the quick-disconnect mount attached, shall not exceed 12.0 pounds.

3.16 SOLDERING. Soldering shall be accomplished in accordance with MIL-S-6872.

3.17 SCREW THREADS. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.18 FINISHES AND PROTECTIVE COATINGS

3.18.1 ALUMINUM ALLOY PARTS. Aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625, except as follows:

3.18.1.1 Small holes and case inserts need not be anodized.

3.18.1.2 Aluminum alloys which do not anodize satisfactory shall be coated with a chemical film in accordance with MIL-C-5541.

3.18.1.3 Where the primary purpose of the treatment is to afford a suitable paint base, chemical treatment in accordance with MIL-C-5541 may be used in lieu of anodizing.

3.18.1.4 Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIL-C-5541 in lieu of anodizing.

3.18.1.5 When abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

3.18.1.6 Parts inclosed in hermetically sealed containers need not be anodized or plated.

3.18.1.7 When the part is plated with tin over a copper flash, the part need not be anodized.

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3.18.1.8 When necessary for electrical bonding, parts need not be anodized.

3.18.2 STEEL PARTS. Where practicable, steel parts shall be cadmium plated in accordance with QQ-P-416, type II or III, as applicable, and of a class that is adequate to achieve the required degree of protection.

3.19 IDENTIFICATION OF PRODUCT. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.20 WORKMANSHIP. The computer, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.20.1 SCREW ASSEMBLIES. Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.20.2 RIVETING. Riveting operations shall be carefully performed to insure that the rivets are tight and satisfactorily headed.

3.20.3 GEARS. Gear assemblies shall be properly aligned and meshed and shall operate without interference, tight spots, loose spots, or other irregularities. Where required for accuracy adjustment, gear assemblies shall be free from backlash.

3.20.4 CLEANING. Before insertion in the case, the computer shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, or other foreign material after assembly. Burrs and sharp edges as well as resin flash that might crumble shall be removed.

4. QUALITY ASSURANCE PROVISIONS

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4.1 RESPONSIBILITY FOR INSPECTION. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS. The inspection and testing of computers shall be classified as follows:

a. Preproduction tests

b. Acceptance tests.

4.3 TEST CONDITIONS

4.3.1 STANDARD ATMOSPHERIC CONDITIONS. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.3.2 ATTITUDE. Unless otherwise specified, the computer shall be tested in normal operating position.

4.3.3 SUPPLY VOLTAGE. Unless otherwise specified, all tests shall be conducted with 115 ±5V ac single phase, 400 ±10 cps.

4.3.4 CONNECTIONS. Whenever it is specified that the computer shall be properly connected, it shall be understood that the computer connector terminals shown on Drawing 64F1866 are connected to a test fixture that provides the required inputs, performs the mode relay excitation connections, and provides the required loads on the output as specified in section 3.

4.3.4.1 TEST FIXTURE INPUTS TO COMPUTER. Inputs from the test fixture to the computer shall simulate the radio signals required by the computer, the pitch and roll signals, the heading and course error signals, and the altitude error signals.

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4.3.4.1.1 TEST FIXTURE LOADS ON COMPUTER. Shunt resistance in series with an external relay shall be provided for the fire-control-mode input. The cables shall be shielded.

4.3.4.1.2 Simulated radio deviation signals shall have manually variable magnitudes from 0 to ±1,000 mv at 0 frequency and 0 to ±100 mv at frequencies of 0.08 to 0.0159 cps where the signal varies sinusoidally. The radio flag signals shall be adjustable from 150 to 285 mv when connected to the computer.

4.3.4.1.3 Roll and pitch signals shall be provided from standard transmitters as specified in 3.11.4. Heading error and course error signals shall be from a control transformer as specified in 3.11.2. The test transmitters shall be manually settable to fixed displacements. The roll and course error signals shall provide 0.08 to 0.0159 cps oscillation when testing the rate circuits.

4.3.4.1.4 The altitude error signal shall be in accordance with 3.11.5.

4.3.4.2 TEST FIXTURE LOADS ON COMPUTER. Two 1,000-ohm loads ±3 percent, in parellel, shall be utilized on each output signal. For each output, the basic load shall be a meter movement in accordance with design criteria specified herein.

4.3.4.3 TEST FIXTURE MODE RELAY. The test fixture shall provide switching so that the various mode relays can be energized individually or in the combinations listed in section 3.

4.3.5 BURN-IN. Burn-in shall be in accordance with MIL-R-26667.

4.4 PREPRODUCTION TESTING

4.4.1 TEST SAMPLES. The test samples shall consist of three computers representative of the production equipment. The samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 TEST REPORT. Three copies of a test report prepared in accordance with MIL-STD-831 shall be furnished to the procuring activity.

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4.4.3 PREPRODUCTION TESTS. The preproduction tests shall consist of all tests specified under 4.6.

4.5 ACCEPTANCE TESTS. Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 INDIVIDUAL TESTS. Each computer shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Leakage

c. Signal adjustments

d. Basic Nav mode

e. Manual heading mode

f. Fire control mode

g. Data link mode

h. Radio track mode

(1) VOR

(2) TACAN

(3) Doppler

i. Altitude hold

j. ILS mode

k. ILS approach

l. Beam sensor

m. Early failure detection.

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4.5.2 SAMPLING PLANS AND TESTS

4.5.2.1 SAMPLING PLAN A. Sample computers shall be selected at random in accordance with the following schedule and subjected to the tests listed below, as described under 4.6:

Production Quantity	No. of Sample Computers
1 to 10	1
11 to 100	2
101 to 200	1
Each additional 200	1 for each 200

a. Individual tests

b. Dielectric strength

c. Low temperature operation

d. High temperature operation

e. Extreme voltage and frequency variation

f. Vibration error

g. Power.

4.5.2.2 SAMPLING PLAN B. Unless otherwise specified, 3 computers shall be selected at random from the first 15 on the contract or order and subjected to the following tests as described under 4.6:

a. Sampling plan A tests

b. Electromagnetic interference

c. Vibration failure

d. High temperature exposure

e. High altitude - low temperature

f. Sand and dust

g. Rain

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h. Fungus

i. Humidity

j. Salt fog

k. Thermal shock

l. Sealing thermal shock

m. Shock

n. Signal characteristics

o. Reliability

p. Acceleration

q. External pressure.

4.5.2.2.1 At the option of the procuring activity and with the exception of sampling plan A tests, tests may be divided into 3 groups, as follows, to be conducted on 3 sets of computers if a need exists to expedite tests. Each computer shall meet the requirements of sampling plan A tests after completing the group of specific tests to which submitted. At the option of the procuring activity, group II may be conducted on empty cases which have all external parts installed or attached as on a completed computer.

Group I	Group II	Group III
Electromagnetic interference	Humidity	Vibration failure
High temperature exposure	Fungus	Acceleration
Low temperature	Rain	Shock
Thermal shock	Salt fog	Reliability
Signal characteristics	Sand and dust	External Pressure
	Sealing thermal shock	

4.5.2.3 SAMPLING PLAN C. Unless otherwise specified, three computers shall be selected at random from the first 20 computers of each production quantity of 100 and subjected to the test specified in 4.6.21.2.

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4.5.2.4 REJECTION AND RETEST. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.4.1 INDIVIDUAL TESTS MAY CONTINUE. For operational and production reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 DEFECTS IN ITEMS ALREADY ACCEPTED. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT. The computer shall be inspected to determine compliance with the requirements herein with respect to materials, workmanship, marking, and design.

4.6.2 LEAKAGE. The computer shall be tested for leakage by means of a mass-spectrometer-type helium leak-detector. The initial maximum detected leak rate, at a pressure differential of one atmosphere, shall not permit more than 10 percent loss of the total filling medium after 1,000 hours.

4.6.3 SIGNAL ADJUSTMENTS. The computer shall be properly connected with all input signals initially set at zero signal or displacement. Each of the adjustments shall be checked for range and specified setting. The values shall be as specified in tables I, II, III, and IV. This test shall not be repeated under sampling plan A or B.

4.6.4 BASIC NAV MODE. The computer shall be properly connected with power applied and with all input signals initially set at zero signal or displacement. Operation of the computer shall be as specified in 3.10.2.1. The computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

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4.6.5 MANUAL HEADING MODE. The computer shall be properly connected with power applied and all input signals initially set at zero signal or displacement. Operation of the computer shall be as specified in 3.10.2.2. The computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

4.6.5.1 FIRE CONTROL MODE. The computer shall be properly connected with power applied and all inputs initially set at zero signal or displacment. No heading error signal shall be applied to the computer during this test. Operation of the computer shall be as specified in 3.10.2.2.1. The computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

4.6.6 DATA LINK MODE. The computer shall be properly connected with power applied and all input signals initially set at zero signal or displacement. Operation of the computer shall be as specified in 3.10.2.3. The computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

4.6.7 RADIO TRACK MODE. The computer shall be properly connected with power applied and all input signals initially set at zero signal or displacement. The three modes of VOR, TACAN, and doppler shall be individually tested under the conditions specified in 3.10.2.4. Computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

4.6.8 ALTITUDE HOLD. The computer shall be properly connected with power applied and all input signals initially set at zero signal or displacement. Operation of the computer shall be as specified in 3.10.2.5. Computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

4.6.9 ILS MODE. The computer shall be properly connected with power applied and all input signals initially set at zero signal or displacement. Operation of the computer shall be as specified in 3.10.2.6. Computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

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TABLE IV

POINTER AND FLAG INDICATIONS

MODE	SUBMODE	VERTICAL STEERING POINTER	HORIZONTAL STEERING POINTER	HSI WARNING FLAG	GLIDESLOPE WARNING FLAG	GLIDESLOPE DISPLACEMENT BAR	VERTICAL POINTER WARNING FLAG
Basic Nav	None	$11^{+2.5}_{-1.0}$ ma (Off Scale Supply)	$11^{+2.5}_{-1.0}$ ma (Off Scale Supply)				255^{+245}_{-0} ua (Off Scale Supply)
Manual Heading	None	Computed Signal					
	Altitude Hold		Computed Signal	255^{+245}_{-0} ua (Off Scale Supply)	255^{+245}_{-0} ua (Off Scale Supply)	500^{+500}_{-0} ua (Off Scale Supply)	255^{+245}_{-0} ua (When Computed Signal is Valid)
	Optional Fire Control	Computed Signal (Sensitive Bank Angle)	$11^{+2.5}_{-1.0}$ ma (Off Scale Supply)				
Data Link	None	Computed Signal					
	Manual Heading	Computed Signal Heading Bank Limit					
	Altitude Hold	Computed Signal	Computed Signal				

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POINTER AND FLAG INDICATIONS

MODE	SUBMODE	VERTICAL STEERING POINTER	HORIZONTAL STEERING POINTER	HSI WARNING FLAG	GLIDESLOPE WARNING FLAG	GLIDESLOPE DISPLACEMENT BAR	VERTICAL POINTER WARNING FLAG
Radio Track VOR	None	Computed Signal	$11^{+2.5}_{-1.0}$ ma (Off Scale Supply)	Connected to VOR/Loc Receiver	255^{+245}_{-0} ua (Off Scale Supply)	500^{+500}_{-0} ua (Off Scale Supply)	255^{+245}_{-0} ua (When Computed Signal is Valid)
	Manual Heading	Computed Signal (Heading Mode)					
	Altitude Hold	Computed Signal (Radio Track VOR)	Computed Signal				
Radio Track TACAN	None	Computed Signal	$11^{+2.5}_{-1.0}$ ma (Off Scale Supply)	Connected to TACAN Receiver			
	Manual Heading	Computed Signal (Heading Mode)					
	Altitude Hold	Computed Signal (Radio Track TACAN)	Computed Signal				
Radio Track Doppler	None	Computed Signal	$11^{+2.5}_{-1.0}$ ma (Off Scale Supply	28V dc on 3J2-CC 255^{+245}_{-0} ua (Off Scale Voltage) 28V dc on 3J2-U Connected to AN/ASN-24(V) and AN/ASN-35 Computer Warning Flag			
	Manual Heading	Computed Signal (Heading Mode)					
	Altitude Hold	Computed Signal (Doppler Mode)	Computed Signal				

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TABLE IV (CONT)

POINTER AND FLAG INDICATIONS

MODE	SUBMODE	VERTICAL STEERING POINTER	HORIZONTAL STEERING POINTER	HSI WARNING FLAG	GLIDESLOPE WARNING FLAG	GLIDESLOPE DISPLACEMENT BAR	VERTICAL POINTER WARNING FLAG
ILS	None	Computed Signal	$11^{+2.5}_{-1.0}$ ma (Off Scale Supply)	Connected to VOR/Loc Receiver	Connected to Glideslope Receiver	Connected to Glideslope Receiver	255^{+245}_{-0} ua (When Computed Signal is Valid)
	Manual Heading	Computed Signal (Heading Mode)					
	Altitude Hold	Computed Signal (ILS Mode)					
ILS APPROACH	None	Computed Signal	Computed Signal				

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4.6.10 ILS APPROACH. The computer shall be properly connected with power applied and all input signals initially set at zero signal or displacement. Operation of the computer shall be as specified in 3.10.2.7. Computer outputs shall not exceed the tolerances for each setting specified in tables I and II.

4.6.10.1 BEAM SENSOR. The computer shall be properly connected with power applied. To simulate tuning of the VOR/Loc radio to a localizer frequency, 28V dc shall be supplied to pin 3J2-n. The computer shall operate as specified in 3.10.3.

4.6.11 EARLY FAILURE DETECTION. This test shall not be repeated under sampling plans A and B. The computer shall be properly connected with power applied and operated for a minimum of 10 hours. All components of the computer shall be energized and operated throughout the test. No failures shall occur. At the completion of this test, the vertical-pointer warningflag circuitry shall be tested and shall meet the requirements of 3.12.2.

4.6.12 DIELECTRIC STRENGTH. A potential of 500V dc shall be applied between isolated pins and between pins and the case for a period of 10 seconds. There shall be no breakdown of insulation.

4.6.13 ENVIRONMENTAL STABILITY. After completion of the following tests (4.6.13.1 through 4.6.13.4.1), the computer shall be subjected to and shall meet the individual tests.

4.6.13.1 LOW TEMPERATURE OPERATION. The computer shall be properly connected, with no power applied, and subjected to a temperature of -54° $+2^{\circ}$ C for a period of 4 hours. At the end of this period and while still at the low temperature, power shall be applied. The computer shall meet all of the individual tests, except sealing. Immediately following removal from the temperature chamber, the computer shall be subjected to the sealing test.

4.6.13.2 HIGH TEMPERATURE OPERATION. The computer shall be properly connected, with no power applied, placed in a chamber, and maintained at a temperature of 71° $+2^{\circ}$ C for a period of 4 hours. While still at the high temperature, power shall be applied and the computer shall meet all individual tests, except sealing. The temperature shall be increased to 98° C for a

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period of 10 minutes with the computer operating. The computer shall meet the requirements of 3.10.2.4 and immediately following removal from the chamber shall meet the test specified in 4.6.2.

4.6.13.3 EXTREME VOLTAGE AND FREQUENCY VARIATION. The computer shall be properly connected and, with input power applied, shall meet the individual tests at all combinations of 105V 380 cps and 125V 420 cps. With input power applied, the computer shall then meet the individual tests at all combinations of 105V 320 cps and 125V 480 cps, except that the tolerances shall be within those specified in tables I and II for extreme voltage and frequency variations.

4.6.13.4 VIBRATION ERROR. The computer shall be properly connected with power applied and subjected to vibration error tests consisting of a frequency survey with vibration applied to the longitudinal, lateral, and vertical axes of the computer. The computer shall be subjected to vibration with a constant applied double amplitude of 0.060 inch through the frequency range of 10 to 55 cps and 10g input from 55 to 500 cps in each of the 3 mutually perpendicular axes. This test may be conducted with vibration applied in a circular motion in a plane 45° to the horizontal plane of the computer. Duration of vibration shall be 1 minute in each axis for 1 complete cycle of frequency range or 1 minute with vibration applied in a circular motion for 1 complete frequency range.

4.6.13.4.1 SIGNAL CONTINUITY. During vibration, there shall be no transients or discontinuity in output for any mode of operation.

4.6.14 POWER CONSUMPTION. The computer shall be properly connected with power applied. The a-c and d-c power consumption for each mode shall not exceed the values specified in 3.13.

4.6.15 ELECTROMAGNETIC INTERFERENCE. The computer shall be tested in accordance with MIL-STD-462, Notice 2, Test Methods CE03, CE04, CS01, CS02, CS06, RE02, RS02, and RS03 to demonstrate compliance with the requirements of paragraph 3.5.1 of this equipment specification.

4.6.16 ENVIRONMENTAL. The computer shall be subjected to the following tests, conducted in accordance with MIL-STD-810 and as specified.

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4.6.16.1 HIGH TEMPERATURE EXPOSURE. This test shall be conducted in accordance with method 501. The period of exposure shall be 24 hours. The computer shall be properly connected, with no power applied during exposure. At the end of the 24-hour period and while still at the high temperature, power shall be applied for 1 hour. The computer shall meet the individual tests specified herein. The computer shall operate without sticking or erratic performance during this test.

4.6.16.2 TEMPERATURE-ALTITUDE. The high altitude-low temperature test shall be conducted in accordance with method 504 for class 2 equipment, except that the altitude shall be 100,000 feet. The computer shall be connected with no power applied. At the end of the 72-hour period, pressure in the chamber shall be reduced to the equivalent of 100,000 +500 feet, and power applied. The computer shall operate immediately and its operation shall be observed for 1 hour. There shall be no evidence of erratic performance. All mode relays and rate circuits shall be tested. Pressure in the test chamber shall be increased to atmospheric and, after the computer has returned to room temperature, it shall meet all individual tests.

4.6.16.3 SAND AND DUST. The sand and dust test shall be conducted in accordance with method 510. Upon completion of this test, the computer shall be examined to determine that no sand or dust has entered the adjustment dust cover. The computer shall then meet the individual tests.

4.6.16.4 HUMIDITY. The computer shall be subjected to a humidity test in accordance with method 507 after which it shall meet the individual tests. There shall be no evidence of corrosion or rust which will affect subsequent operation.

4.6.16.5 FUNGUS. The fungus test shall be conducted in accordance with method 508, proceure I. Upon completion of this test, the computer shall meet the individual tests. There shall be no deterioration nor shall any part of the computer support fungus growth.

4.6.16.6 RAIN. The rain test shall be conducted in accordance with method 506 in its normal operating position. The computer shall then meet the individual tests.

4.6.16.7 SALT FOG. The computer shall be subjected to a salt fog test in accordance with method 509. At the end of the 48-hour period, the computer shall be subjected to and shall meet the individual tests.

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4.6.16.8 VIBRATION FAILURE. The computer shall be properly connected with power applied and subjected to a vibration test in accordance with method 514, 1B1C. At the end of this test, the computer shall be subjected to and shall meet the individual tests. The computer shall then be subjected to a vibration of 0.4 inch double amplitude or 1.0g (whichever is less) within the frequency range of 1 cps to 10 cps for a period of 15 minutes. There shall be no transients or discontinuity in outputs for any mode of operation. No damage to the computer shall occur as a result of this test.

4.6.17 ACCELERATION. The computer, not operating, shall be mounted on a centrifuge in its normal operating position and subjected to an acceleration of 20g for a period of 1 minute in each axis, first along the vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. At the end of this acceleration, the computer shall meet the individual tests. No damage to the computer shall result from this test.

4.6.18 TEMPERATURE SHOCK. The computer shall be subjected to a temperature shock test in accordance with method 503. After the computer has returned to room temperature, connectors and seals shall be examined. There shall be no evidence of cracked terminals or leaks in the seals. The computer shall then meet the individual tests.

4.6.19 SEALING TEMPERATURE SHOCK. The computer shall be immersed alternately in tap water maintained at 85° ±4° C and 5° ±4°C for a total of 8 cycles. The length of time for each bath immersion shall be 30 minutes and no more than 5 seconds shall elapse between bath immersions. The forward 1/2 inch of the front adjsutment panel need not be submerged. No damage shall result to the hermetic seal as a result of this test. Following the immersions; the computer shall be tested for leaks by means of a mass-spectrometer-type helium leak detector. The initial maximum detected leak rate, at a pressure differential of 1 atmosphere, shall not permit more than 10 percent loss of the total filling medium after 1,000 hours.

4.6.19.1 EXTERNAL PRESSURE. With no power applied, the computer shall be cycled 6 times from atmospheric pressure to 26.5 psia. At the completion of this test, the computer shall pass all the individual tests.

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4.6.20 SHOCK. The computer shall be subjected to a shock test in accordance with method 516, procedure I. The equipment shall not be damaged or subsequently fail as a result of this test.

4.6.21 RELIABILITY.

4.6.21.1 Three (3) computers shall be subjected to a reliability demonstration test in accordance with MIL-STD-781B, Test Plan II, which shall demonstrate a computer minimum acceptable MTBF of 600 hours. The test environment shall be in accordance with MIL-STD-781B, Test Level E. Each mode covered in paragraph 3.10.2.1 through 3.10.2.7 shall be switched at 1 hour intervals during the ON time of this test. All inputs shall be applied to the computer at their maximum value during the test. The output loads shall be 500 ohms.

4.6.21.2 An acceptance reliability test shall be conducted to demonstrate a computer minimum acceptable MTBF of 600 hours. Acceptance shall be based on MIL-STD-781B, Test Plan XXVII and the test environment of Test Level E. Eachmmode covered in 3.10.2.1 through 3.10.2.7 shall be switched at 1 hour intervals during the ON time of this test. All inputs shall be applied to the computer at their maximum value during test. The outputs shall be 500 ohms.

4.6.22 SIGNAL CHARACTERISTICS

4.6.22.1 Vertical and horizontal output signal characteristics shall be tested to determine that the requirements of 3.12 and 3.12.1 are fulfilled.

4.6.22.2 Computer loading of input signals shall be tested. Input impedance for attitude signals, heading and course error signals, and radio signals shall be as specified in 3.11.

4.7 PACKAGING INSPECTION

4.7.1 QUALITY CONFORMANCE. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with thequality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

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4.7.2 FIRST ARTICLE. When the unit container is capable of serving as the shipping container First Article Inspection, and rough handling tests as outlined in section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to insure freedom from operational malfunction.

5. PACKAGING

5.1 PRESERVATION-PACKAGING. Preservation-packaging shall be level A, B, or C as specified by the procuring activity.

5.1.1 LEVEL A.

5.1.1.1 CLEANING. Unit shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 DRYING. Immediately after cleaning, the unit shall be dried following any one or combination of the drying procedures of MIL-P-116. The drying procedures employed shall not be injurious to the unit.

5.1.1.3 PRESERVATION APPLICATION. Not required.

5.1.1.4 UNIT PACKAGING. Unless otherwise specified by the procuring activity, each unit shall be packaged in quantity unit packs of one each in accordance with Method IAB of MIL-P-116. Overbox each completed pack in a PPP-B-636 Fiberboard, weather-resistant class box. Cushioning material between bag and container shall be of a type, density and thickness that will protect the unit from impact shock damage when tested as specified in paragraph 4.7.2 of this specification.

5.1.2 LEVEL B. Each unit shall be packaged as indicated for Level A except that a specification PPP-B-636 class domestic box shall be used as the unit container.

5.1.3 LEVEL C. Units shall be cleaned, dried, and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 PACKING. Packing shall be level A, B, or C as specified (see 6.2).

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5.2.1 LEVEL A. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636 weather-resistant. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, specification PPP-B-601 plywood box, shall be used for the shipping container.

5.2.2 LEVEL B. Units packaged as specified in 5.1 shall be packed in shipping containers conforming to specification PPP-B-636, class domestic. Closure and reinforcing requirements shall be in accordance with the appendix of PPP-B-636. For the Army, the shipping containers shall conform to specification PPP-B-636 class weather resistant.

5.2.3 LEVEL C. Packaged units shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct domestic shipment from the supply source to the first receiving activity. These packs shall conform to applicable carrier rules and regulations.

5.3 MARKING. In addition to any special marking required by the contract or order, each unit package and shipping container shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows: Computer, Flight Director CPU-65/A.

6. NOTES

6.1 INTENDED USE. The CPU-65/A computer covered by this specification is intended for universal use in all types of military fixed-wing aircraft, such as cargo, bomber, and high-performance fighters, to provide computed flight director signals to flight director indicators.

6.2 ORDERING DATA. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B and C tests will not be conducted

c. Selection of applicable levels of packaging and packing

d. Reinspection date markings.

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6.3 DEFINITION

6.3.1 HERMETIC SEAL. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic, or both materials. A hermetic seal is not intended to include seals accomplished by gaskets.

Custodian:
Air Force - 99

Preparing Activity:
Air Force - 99

Project Number:
6610-F271

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NOTICE OF INACTIVATION
FOR NEW DESIGN

INCH-POUND

MIL-C-38286B(USAF)
NOTICE 1
15 JANUARY 1996

MILITARY SPECIFICATION

COMPUTER, FLIGHT DIRECTOR CPU-65/A

This notice should be filed in front of MIL-C-38286B(USAF) dated 16 May 1990.

MIL-C-38286B(USAF) is inactive for new design and is no longer used, except for replacement purposes.

Custodians:
Air Force - 99

Preparing Activity:
Air Force - 71

Agent:
Air Force - 99

(Project 6610-F423)

AMSC N/A FSC 6610

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INCH-POUND

NOTICE
OF CANCELLATION

MIL-C-38295A(USAF)
NOTICE 1
27 May 90

MILITARY SPECIFICATION

COMPUTER SET, ATTITUDE-REFERENCE
BOMBING AN/AJB-7

MIL-C-38295A(USAF), dated 28 August 1968, is hereby cancelled without replacement.

Preparing Activity:
AIR FORCE -71

(Project 1280-F051)

AMSC N/A

FSC 1280

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NOTICE
OF CANCELLATION

MIL-C-38303A(USAF)
NOTICE 1
12 August 1988

MILITARY SPECIFICATION

CONTROLLER, COMPASS SYSTEM

MIL-C-38303A(USAF), dated 5 October 1970, is hereby canceled without replacement.

Preparing activity:
AIR FORCE -71

Project number:
6615-F262

AMSC N/A

FSC 6615

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NOTICE
OF CANCELLATION

INCH-POUND

MIL-C-38304(USAF)
NOTICE 1
2 Mar 90

MILITARY SPECIFICATION

COMPUTER, BOMB RELEASE, ANGLE CP-735/AJB-7

MIL-C-38304 (USAF), 31 July 1964, is hereby cancelled without replacement.

Preparing Activity:
AIR FORCE -71

Project No. 1280-F050

AMSC N/A

FSC 1280

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NOTICE
OF CANCELLATION

INCH-POUND

MIL-C-38333(USAF)
NOTICE 1
3 November 1989

MILITARY SPECIFICATION

CONTROLLER, COMPASS SYSTEM
C-6448/ASN-55

MIL-C-38333 (USAF), dated 3 August 1964, is hereby cancelled without replacement.

Preparing Activity:
AIR FORCE -71

(Project 6615-F290)

AMSC N/A

FSC 6615

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NOTICE OF
CANCELLATION

MIL-C-38351A(USAF)
NOTICE 1
18 JULY 1988

MILITARY SPECIFICATION

CLOTH, PARACHUTE, NYLON, AROMATIC, NONMELTING

Military specification MIL-C-38351A(USAF) dated 22 August 1972 is hereby canceled.

Preparing activity:
Air Force - 11

Project 8305-F251

FSC 8305

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MIL-C-38392A(USAF)
11 June 1969
Superseding
MIL-C-38392 (USAF)
18 June 1965

MILITARY SPECIFICATION

COMPRESSORS, HERMETIC, AIR CONDITIONING, GENERAL SPECIFICATION FOR

1. SCOPE

1.1 Scope. This specification covers the requirements for 50/60-Hz and 400-Hz, electric-motor-driven hermetic compressors for use with air conditioni equipment using refrigerant -22 (monochlorodifluoromethane) and as specified in MIL-A-38339.

1.2 Classification. Compressors shall be of the following classes and types, as specified (see 6.2):

Class 1	- 7,000 BTU/hr nominal capacity
Type I	- 115-V, 50/60-Hz, single-phase
Type II	- 208-V, 400-Hz, 3-phase
Class 2	- 11,000 BTU/hr nominal capacity
Type I	- 115-V, 50/60-Hz, single-phase
Type II	- 230-V, 50/60-Hz, single-phase
Type III	- 208-V, 50/60-Hz, 3-phase
Type IV	- 208-V, 400-Hz, 3-phase
Class 3	- 20,000 BTU/hr nominal capacity
Type I	- 230-V, 50/60-Hz, single-phase
Type II	- 208-V, 50/60-Hz, 3-phase
Type III	- 208-V, 400-Hz, 3-phase
Class 4	- 40,000 BTU/hr nominal capacity
Type I	- 208-V, 50/60-Hz, 3-phase
Type II	- 208-V, 400-Hz, 3-phase
Class 5	- 60,000 BTU/hr nominal capacity
Type I	- 208-V, 50/60-Hz, 3-phase
Type II	- 208-V, 400-Hz, 3-phase

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	Box, Wood, Cleated-Plywood
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-P-116	Preservation, Methods of
MIL-A-38339	Air Conditioners, Lightweight, Compact, Military, General Requirements For

STANDARDS

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods
MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing, with Appropriate Test Methods
MS3102	Connector, Receptacle, Electric, Box Mounting

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society of Heating, Refrigeration, and Air Conditioning Engineers, Incorporated

ASHRAE Standard 23	Methods of Testing and Rating Refrigerant Compressors

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(Application for copies should be addressed to the American Society of Heating, Refrigeration, and Air Conditioning Engineers, Incorporated, 345 East 47th Street, New York, New York 10017.)

National Electrical Manufacturers Association

Standard MG-1 Standards For Motors and Generators

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

Underwriters' Laboratories, Incorporated

UL Std No. 303 Condensing Units, Refrigeration

(Application for copies should be addressed to the Underwriters' Laboratories, Incorporated, 161 Sixth Avenue, New York, New York 10013; 207 East Ohio Street, Chicago, Illinois 60611; or 1655 Scott Boulevard, Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Materials. Exposed metal surfaces of the compressors shall be of the corrosion-resistant type or treated to resist corrosion due to atmospheric conditions likely to be met in storage and normal service and shall be suitable for the purpose. Materials shall be free from defects and imperfections that might affect the serviceability of the finished compressors.

3.3 Design

3.3.1 Reliability. The compressors shall each have a mean-time-between-failure (MTBF) of not less than 1,300 hours at a confidence factor of 0.90 (see 6.3.1 and 6.3.2).

3.3.2 Operating life. The compressors shall be capable of operating for 2,740 hours without loss of efficiency during that period.

3.3.3 Tilted position operation. The compressors shall be capable of operation without failure when tilted 10 degrees from vertical in any direction.

3.3.4 Refrigerant. Compressors shall be suitable for operation with refrigerant -22 (monochlorodifluoromethane).

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3.3.5 Motor cooling. The compressor motor shall be refrigerant-, air-, or oil-cooled, or any combination of the three.

3.3.6 Rupture pressure. The compressors shall be designed to withstand pressures up to 675 psig in both the suction and discharge cavities without rupture.

3.3.7 Compressor oil. The compressors shall be filled with lubricating oil to the level recommended by the manufacturer. The compressors shall be supplied with a grade and type of lubricating oil as recommended by the compressor manufacturer.

3.3.8 Housing. The compressor and compressor motor shall be enclosed in the same housing with the mounting hardware and no external shaft or shaft seal, and with the motor running in a refrigerant atmosphere. The housing may be of either the fully welded or brazed type, or of the service sealed type. In the fully welded or brazed type, the housing shall be permanently sealed and shall not be provided with means of access for servicing in the field. In the service-sealed type, the housing shall be provided with some means of access for servicing internal parts.

3.4 Performance

3.4.1 Operation. The compressors shall be capable of continuous operation under the conditions specified in table I.

TABLE I. Operating Conditions

Saturated Discharge Temp. (^{o}F)	Saturated Suction Temp. (^{o}F)	Ambient Temp. (^{o}F)
160	40 to 60	160
150	30 to 60	150
140	30 to 60	140
130	30 to 60	130
120	30 to 40	120
60	-20 to 20	-20 to 60

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3.4.1.1 Classes 1, 2, and 3 compressors shall be capable of operation in ambient temperatures down to 0°F. Classes 4 and 5 compressors shall be capable of operation in ambient temperatures down to -25°F. These requirements shall be obtained with the compressor installed in the appropriate air conditioner (MIL-A-38339).

3.4.2 Temperature conditions. The lubrication and overload selection shall permit operation of the compressors without damage in ambient temperatures ranging from 0° to +160°F for classes 1, 2, and 3 compressors, and from -25° to +160°F for classes 4 and 5 compressors. All compressors shall be capable of withstanding storage temperatures ranging from -65° to +160°F. This compressor suction cavity shall be capable of withstanding without damage, instantaneous pressure changes from the pressures corresponding to the minimum saturated evaporator temperature to the maximum saturated discharge temperature specified in 3.4.1.

3.4.3 Capacity and power requirements. The compressors specified herein shall have the capacity and power requirements specified in table II.

TABLE II. Compressor Capacities and Power Requirements

CLASS	TYPE	CAPACITY (MINIMUM)	BTU/HR (MAXIMUM)	BTU/WATT/HR (MINIMUM)	POWER FACTOR (MINIMUM)	ALLOW. POWER (KW) (MAXIMUM)
1	I	7,400	8,600	5.7	.85	1.4
	II	7,400	8,600	5.3	.59	1.6
2	I	11,000	13,100	5.4	.85	2.3
	II	11,000	13,100	5.8	.85	2.1
	III	11,000	13,100	6.0	.85	2.0
	IV	11,000	13,100	5.5	.60	2.2
3	I	22,200	26,000	6.0	.85	4.2
	II	22,200	26,000	6.0	.85	4.2
	III	22,200	26,000	5.7	.60	4.4
4	I	41,500	49,000	5.9	.85	7.5
	II	41,500	49,000	5.7	.60	8.0
5	I	65,000	76,000	5.9	.85	12.0
	II	65,000	76,000	5.7	.60	13.0

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3.4.3.1 The ratings specified in table II shall apply with the compressor operating at the following conditions:

Ambient air temperature blowing over compressor at approximately

600 fpm	150°F
Saturated discharge temperature	150°F
Evaporator temperature	50°F
Subcooling temperature	10°F
Suction superheat	15°F

3.4.4 Low temperature operation. Crankcase heaters shall be provided on classes 1, 2, and 3 compressors if required to prevent undesirable migration of liquid refrigerant to the compressor during shutdown or high starting torque conditions. Classes 4 and 5 compressors shall be provided with crankcase heaters to permit starting at -25°F ambient conditions.

3.4.5 Electric motors. The motor rated voltage phase and frequency shall be as specified in 1.2.

3.4.5.1 Starting torque. The starting torque requirements of the compressors shall be low, as the compressor will always start with equalized high and low side pressures. Kits for reluctance starting (run capacitor, start capacitor, and switching relay) shall be provided on all dual 50/60-Hz, single-phase compressors when starting torque is marginal.

3.4.5.2 Variation from rated voltage. The compressor shall start and operate as specified in NEMA Standard MG-1, paragraph entitled Variation From Rated Voltage, with voltage variations as specified for induction motors.

3.4.5.3 Variation from rated frequency. The compressor shall start and operate as specified in NEMA Standard MG-1, paragraph entitled Variation From Rated Frequency.

3.4.5.4 Combined variation of voltage and frequency. The compressor shall start and operate as specified in NEMA Standard MG-1, paragraph entitled Combined Variation Of Voltage And Frequency, except the combined variation tests shall be conducted with the compressor operating at rated conditions as specified in 3.4.3 herein.

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3.4.5.5 Motor protection (see 6.3.3 and 6.3.4). The compressor motor shall be equipped with a combination protector. The current protector shall be located remotely from the compressor and shall sense the current in all ungrounded power leads of the 3-phase compressor and at least one current-carrying line in single-phase compressors. The current sensor shall be required to break the above current-carrying lines upon current overload. In addition, the current-sensing devices shall contain a set of auxiliary contacts connected into a direct-current control circuit for the air conditioner. The auxiliary contacts shall open with the current sensor on compressor current overload; thereby interrupting the direct-current control circuit and breaking all current-carrying lines to the compressor. The auxiliary contacts shall reset automatically when the current sensor is manually reset. The auxiliary contacts shall be sized for compatibility with the control circuit. The current-sensing device shall have inherent temperature-compensation features and shall be adequately sized to prevent nuisance failures. A moistureproof, transparent, flexible seal shall be provided over the reset switch and shall in no way restrict the movement or prevent resetting the switch.

3.4.5.6 Electrical connections. Each compressor shall be provided with an electrical connector of the lightweight, environmental MS series. Connections shall conform to the wiring diagrams of the specific air conditioner (see 1.2). The MS connectors shall be as follows:

Compressor Class	Connector
1	MS3102-16S-1P
2 and 3	MS3102-20-15P
4 and 5	MS3102-24-11P

3.4.6 Environmental exposure. The compressors shall conform to all requirements specified herein after being exposed to the following environmental conditions:

a. Salt-water sea atmosphere

b. Fungus growth as encountered in tropical climates

c. Sand and dust particles as encountered in desert areas

d. Rain as encountered in any locale

e. Humidity.

3.4.6.1 Vibration. The compressors shall successfully operate after being subjected to vibration as specified in test method 514 of MIL-STD-810 for category h equipment.

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3.4.6.1.1 Vibration transmittance. The compressors shall be so designed that less than +0.1g acceleration (at any frequency) will be transmitted to the adjoining structure.

3.4.6.2 Shock. The compressors shall satisfactorily operate without impaired performance or breakage after being subjected to shock as specified in procedures I and II of method 516 of MIL-STD-810.

3.4.7 Weight, dimensions, connection sizes, and mounting. The weight of each compressor shall not exceed the requirements specified in table III. (The compressor plan view projection dimensions and heights are overall, and are the maximum allowable.) The compressors shall be provided with sweat-type refrigerant suction and discharge fittings suitable for use with copper tubing, and of the sizes listed in table III. A minimum length corresponding to the tubing diameter shall be provided for attachment of copper tubing to the mating sweat fitting. Mounting dimensions shall be as specified in table III, and the specific air conditioner as listed in MIL-A-38339 (see 1.2). The location and orientation of connections, overall configurations, and mounting shall permit interchangeability of alternate compressors.

TABLE III

CLASS	TYPE	MTG DIM (INCHES-NOM)	SUCTION CONN (INCHES-NOM) IDF / ODM	DISCH CONN (INCHES-NOM) IDF / ODM	MAX COMPL WT. (LBS[1])	MAX PROJ. VIEW PROJ. DIM (INCHES)	MAX OA HEIGHT (INCHES)
1	All	4.81X8.00	0.500	0.375	50	9.25X9.75	12.25
2	All	4.81X8.00	0.500	0.375	50	9.25X9.75	12.25
3	I and II	7.50X7.50	0.750	0.500	87	10.50X12.50	16.56
	III	7.50X7.50	0.750	0.500	76	10.50X12.50	16.56
4	I	7.50X7.50	0.875	0.625	100	13.50X12.88	17.00
	II	7.50X7.50	0.875	0.625	93	13.50X12.88	17.00
5	I	8.50X8.50	1.125	0.625	125	14.75X12.75	17.50
	II	8.50X8.50	1.125	0.625	120	14.75X12.75	17.50

[1] Compressor weight shall include complete compressor with oil charge, oil separator, and an oil cooler (if required).

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3.5 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.6 Electromagnetic interference. The compressors shall conform to the electromagnetic interference requirements as specified for class Am equipment in MIL-STD-826 when assembled into the air conditioner.

3.7 Acoustic noise level. The noise level of the hermetic compressors when operating in the applicable air conditioner (MIL-A-38339) shall not cause the air conditioner to exceed the noise level limits specified in the applicable detail specification when tested in accordance with the noise level test of MIL-A-38339.

3.8 Operational markings

3.8.1 Wiring diagram. A wiring diagram which incorporates all circuits from the power source to the motor shall be furnished. This diagram may be affixed to the compressor on the inside of the control box. If affixed to the compressor, it shall be inscribed on a corrosion-resistant plate.

3.8.2 Nameplate data. The following data shall be incorporated in a nameplate made of corrosion-resistant material or corrosion-protected material and securely attached to the compressor in a conspicuous place:

Refrigerant _____
Manufacturer's name _____
Manufacturer's part No. _____
Serial No. _____
Specification No. _____
Class - Type _____
Volts, Phase _____
Frequency (Hz) _____
Full load current _____ (Amps)
Lock Rotor Current _____ (Amps)
Ult. Trip Current _____ (Amps) (71°C)
Model of Motor Protectors _____
Contract No. _____
Type and Recommended Oil _____
Oil Charge - (Pints) _____

3.9 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

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3.10 Workmanship. The compressors shall be fabricated, assembled, and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from defects, burrs, and sharp edges; accuracy of dimensions, thoroughness of soldering, welding, brazing, and riveting; and alignment of parts and tightness of assembly screws, bolts, et cetera.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the compressors shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions. The test conditions shall be as described under the tests to which they apply.

4.3.1 Readings. Readings of all pertinent pressures, temperatures, rotational speeds, and other necessary data required to demonstrate compliance with applicable portions of section 3 shall be recorded at intervals not exceeding 30 minutes during all tests specified herein.

4.3.2 Post-test inspection. Upon completion of all testing or when deemed necessary by the contracting officer, the compressor may be removed from the shell of a welded assembly or the covers removed on a field service type for physical examination.

4.4 Preproduction testing (see 6.2)

4.4.1 Test samples. The test samples shall consist of models representative of the production equipment. Unless otherwise specified (see 6.2), two prototypes of each type compressor required by the contract shall be subjected to the preproduction tests specified in 4.4.3.

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4.4.2 Test report. Upon completion of the preproduction tests, a preproduction test report in accordance with MIL-STD-831 shall be prepared and two complete copies and one reproducible copy of the report furnished to the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be submitted as an attachment accompanying the preproduction test report (same quality and quantity of copies) or shall be included as part of that report:

a. All failures, maintenance, and other events recorded shall be identified by accumulated operating time, miles, cycles, or position in the test procedure as appropriate. Test conditions during the failures or irregular operations identified shall be recorded.

b. Summary of the engineering analysis and of any tests conducted to determine assignable cause for all failures and irregular operations identified.

c. Summary of the engineering analysis behind any corrections made, to be made on production items, or proposed to be made and behind the predicted effectiveness of those corrections.

d. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity or contractor comments on use or maintenance conditions or procedures to be avoided or cultivated to increase the reliability and useful life of the item.

4.4.2.2 Preproduction test approval. All compressors that have been successfully tested and have had preproduction test approval to this specification must have the compressor motors identified by the manufacturer's name and part number as it appears on the motor. Any preproduction test approval given to a particular compressor shall apply only for the motor part number used during the original preproduction testing. The same compressor with a different motor shall have to be retested in accordance with the preproduction tests of this specification to be accepted.

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests specified under 4.6. When two samples are used, they shall be allocated to the preproduction tests as follows and tested as described under 4.6:

Sample No. 1

Examination of product	See 4.6.1
Storage test	See 4.6.3
Environmental tests	See 4.6.4

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High potential test	See 4.6.5
Capacity tests	See 4.6.6
Vibration transmittance test	See 4.6.12
Vibration test	See 4.6.13
Shock test	See 4.6.14
Endurance test	See 4.6.16.

Sample No. 2

Examination of product	See 4.6.1
Tilted position test	See 4.6.9
Winding temperature test	See 4.6.7
Locked rotor test.	See 4.6.8
Electromagnetic interference test	See 4.6.11
Noise level test	See 4.6.10
Pressure tests	See 4.6.15.

4.5 Acceptance tests. The acceptance tests shall consist of the individual tests.

4.5.1 Individual tests. Each compressor shall be subjected to the following tests as described under 4.6:

a. Examination of product	See 4.6.1
b. Capacity verification (except that the test run shall be of 30 minutes duration)	See 4.6.2

4.6 Test methods

4.6.1 Examination of product. The compressor shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, markings, and finishes.

4.6.2 Capacity verification test. A capacity test at 50°F saturated suction temperature, 150°F saturated discharge temperature, and 65°F actual suction temperature shall be conducted in accordance with 4.6.6. Test runs shall be of 30 minutes duration. The resistance to ground shall be recorded. Winding temperatures in excess of motor manufacturer's guaranteed limits or automatic trip-points at less than motor manufacturer's limits or for reasons other than motor protection shall be cause for rejection.

4.6.3 Storage tests

4.6.3.1 High temperature test. The compressor (with normal operating oil and refrigerant -22 charge) shall be subjected to the high temperature test specified in procedure I of MIL-STD-810, method 501. Both the high- and

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low-pressure cavities of the compressor shall be subjected to liquid refrigerant -22 pressures corresponding to the saturated temperatures throughout exposure. The compressor shall be satisfactorily operated upon conclusion of this test.

4.6.3.2 Low temperature test. The compressor shall be subjected to the low temperature test specified in procedure I of MIL-STD-810, method 502, except the exposure temperature shall be -65°F and maintained for a period of not less than 48 hours. The compressor shall be satisfactorily operated upon conclusion of this test.

4.6.4 Environmental tests. Upon completion of the storage tests specified in 4.6.3.1 and 4.6.3.2, the following tests shall be conducted in accordance with the specified procedures of MIL-STD-810:

Test	Procedure
Sand and dust	Procedure I, method 510
Salt-fog	Procedure I, method 509
Fungus resistance	Procedure I, method 508
Rain	Procedure I, method 506
Humidity	Procedure I, method 507

4.6.5 High potential test. The compressor electrical circuit shall be subjected to and shall satisfactorily complete a high potential test in accordance with NEMA Standard MG-1, paragraph entitled High Potential Test.

4.6.6 Capacity tests. Capacity tests shall be conducted in accordance with ASHRAE Standard 23, as modified to meet the temperatures and pressures specified herein and shall be run prior to the endurance test. The results of capacity and wattage input from the operational tests specified in 4.6.6.2 (runs 1 and 10) shall not vary more than 3 percent. In addition to the data required by 4.3.1, the following shall be recorded:

a. Capacity (BTU/hr)

b. Voltage and frequency

c. Amperes

d. Watts input

e. Power factor

f. Winding temperature (resistance method of Seely circuit tester).

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4.6.6.1 Test conditions. The following conditions shall apply for all capacity tests:

a. Voltage variation (±2V)

b. Frequency variations (±1/2 cycle)

c. Subcooling (10°F)

d. Superheat at compressor (15°F)

e. Ambient airflow over compressor (600 fpm).

4.6.6.2 Operational tests. Capacity test runs shall be performed under the conditions specified in table IV and shall be conducted in sequential order from 1 to 10. The compressor shall be mounted in a manner simulating that under actual service. Each test run shall be of 5 hours duration.

TABLE IV. Capacity Test Run Conditions

Run	Saturated Discharge Temp °F	Ambient Temp °F	Saturated Suct Temp °F
1	120	120	30
2	130	130	30
3	130	130	60
4	150	150	30
5	150	150	50
6	150	150	60
7	160	160	40
8	160	160	60
9	60	20	20
10	120	120	30

4.6.7 Winding temperature test. The winding temperatures shall be measured by the resistance method. The compressor shall be operated at 150°F saturated discharge temperature, 50°F saturated suction temperature, and 150°F ambient air temperature across the compressor. The temperature of the suction gas entering the compressor shall be increased in 5°F increments starting at 60°F. The winding temperature shall be recorded for each suction gas temperature increment. The period of operation for each increment shall be of

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sufficient length for stabilized winding temperature. The test shall be continued until the maximum winding temperature, as recommended by the manufacturer, is reached or the compressor overtemperature protector opens the power circuit. The foregoing tests shall be conducted at rated voltage and frequency.

4.6.8 Locked rotor test. The locked rotor test shall be conducted in accordance with Underwriters' Laboratories, Incorporated Standard No. 303 paragraph entitled Locked Rotor Test - Hermetic Compressor Motors.

4.6.9 Tilted position test. The compressor shall be operated while tilted 10 degrees from the horizontal position in each of the following directions: front, back, and each side. The compressor shall operate for 4 hours at rated conditions in accordance with 4.6.6 in each of the four specified positions.

4.6.10 Acoustic noise level test. The noise levels produced by the appropriate air conditioner, when the applicable compressor of the detail specification is installed, shall not exceed the sound pressure level limits of the detail specification when tested in accordance with the noise level test of MIL-A-38339.

4.6.11 Electromagnetic interference test. When installed in the air conditioner, the compressor shall assure compliance with the applicable electromagnetic interference test requirements of MIL-STD-826, class Am, as follows: Test methods 1001, 2001, 3001, 3002, and 4001. Short duration (transient) interference requirements of test methods 3002 and 4001, are relaxed 20 db for the air conditioners.

4.6.12 Vibration transmittance test. A vibration survey of each test run of 4.6.6.2 shall be conducted. Transmitted acceleration (at any frequency) in excess of $\pm$0.1g shall be cause for rejection.

4.6.13 Vibration test. A compressor mounting bracket shall be fabricated and attached to a rigid base. The compressor shall be mounted on the bracket using 70 durometer vibration washers similar to those covered by Industrial Rubber Goods Company part No. D-4525-265. The compressor shall satisfactorily operate after being subjected to a vibration test conducted in accordance with MIL-STD-810, test method 514, equipment category h, procedure X and curve AB. The resonant search and resonant dwell shall be at 1-1/4g input, and the sinusoidal cycling test shall be at 2-1/2g input. The time used for resonant search shall not be counted in total test time.

4.6.14 Shock test. The compressor shall satisfactorily operate after being subjected to a shock test conducted in accordance with MIL-STD-810, method 516, procedure I with a shock wave in accordance with figure 516-1 (terminal-peak sawtooth) with an amplitude of (b) and a time duration of (d) as listed for ground equipment, and procedure II except that the compressor shall be

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dropped on each face and each corner and each edge from a height of 30 inches for class 1 through class 4 compressors and a height of 21 inches for the class 5 compressor.

4.6.15 Pressure tests. Pressure tests shall be performed on and applied internally to the compressor crankcase under the conditions specified in table V. The capacity verification test specified in 4.6.2 shall be conducted after both gas tests, except that the test run shall be of 30 minutes duration. Failure to pass the capacity verification test or rupture at pressure less than 675 psig shall constitute failure of this test.

TABLE V. Conditions For Pressure Tests

Test	Cycles	Pressure Range (PSIG)	Minimum Pressure Increase Time Rate
Gas	15	0 to 434	5 pounds/second
Gas	24	0 to 380	Instantaneous
Rupture (Hydrostatic)	1	0 to 675	5 pounds/second

4.6.16 Endurance test. An endurance test shall be conducted on compressors which have successfully passed the tests specified in 4.6.6. The airflow across the compressor during the endurance testing shall be approximately 600 fpm. The ambient temperature throughout the endurance test shall be the same as the specified saturated discharge temperatures. The refrigerant oil shall withstand the discharge temperatures of the compressor for the duration of the life test.

4.6.16.1 Test schedule. The schedule specified in table VI shall be followed in performing the endurance tests. Resistance to ground shall be recorded after each 100 hours of operation or each test, whichever is the less time.

4.6.16.2 Test loop. The compressor shall be tested on a refrigerating test loop of sufficient sizing for each class of compressor tested.

4.6.16.3 Test installation. When installed in the appropriate air conditioner (MIL-A-38339), the compressors shall satisfactorily pass the Endurance tests of MIL-A-38339, except for the Heating test.

4.6.17 Reliability demonstration. Satisfactory completion of all tests specified herein without failure demonstrates compliance with the quantitative reliability requirements.

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TABLE VI. Endurance Test Schedule

RUN 1. 250 hours of intermittent operation under the following conditions:

15 minutes on
5 minutes off

a. 150°F saturated discharge temperature

b. Suction gas temperature, 25°F superheat

c. 150°F ambient temperature

d. Evaporation temperature shall be varied periodically with approximately one-third of the total 250 hours being spent at each of the saturated suction temperatures of 40°, 50°, and 60°F.

RUN	HOURS	VOLTS	HERTZ	SATURATED SUCTION TEMP °F	SATURATED DISCHARGE TEMP °F	ACTUAL SUCTION TEMP °F
2	16	Low (5%)	Rated	50	150	75
3	16	High (10%)	Rated	50	150	75
4	16	Rated	Low (5%)	50	150	75
5	16	Rated	High (5%)	50	150	75
6	16	Low (5%)	Rated	40	150	65
7	16	High (10%)	Rated	40	150	65
8	16	Rated	Low (5%)	40	150	65
9	16	Rated	High (5%)	40	150	65
10	16	Low (5%)	Rated	45	160	70
11	16	High (10%)	Rated	45	160	70
12	140	Rated	Rated	35	150	60
13	140	Rated	Rated	40	150	65
14	150	Rated	Rated	45	150	70
15	150	Rated	Rated	50	150	75
16	500	Rated	Rated	30	100	55
17	500	Rated	Rated	40	125	65
18	500	Rated	Rated	50	150	75
19	250	Rated	Rated	60	160	85
20	250	Rated	Rated	20	60	45

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4.7 Inspection of the preservation, packaging, packing, and marking for shipment and storage. The inspection of the preservation, packaging, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall level A or C as specified (see 6.2).

5.1.1 Level A. Compressors shall be preserved and packaged in accordance with method III of MIL-P-116. All openings of the electrical connection box shall be sealed with class 1 tape conforming to PPP-T-60.

5.1.2 Level C. Preservation and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Each compressor, preserved and packaged as specified in 5.1.1, shall be packed in an overseas-type style B container conforming to PPP-B-601. Containers shall be of uniform shape and size and minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-601.

5.2.2 Level B. Level B packing shall be the same as specified for level A, except that containers shall be of the domestic type, style A.

5.2.3 Level C. Compressors, preserved and packaged as specified in 5.1.2, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

5.3 Physical protection. Internal blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. The free-fall drop test shall be in accordance with appendix A of MIL-STD-1186. The test shall not be required when level C packing is specified.

5.4 Marking. In addition to any special marking required by the contract or order (see 6.2), the interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The hermetic compressors covered by this specification are for use in combination with other standard refrigeration components in air conditioning systems which employ refrigerant-22.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Class and type of compressor required (see 1.2)

c. Point of inspection and quantity of preproduction samples (see 4.4 and 4.4.1)

d. Selection of applicable levels of preservation, packaging, and packing (see 5.1 and 5.2)

e. Special shipment marking (see 5.3)

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

Confidence Level = 90 Percent

$$\text{MTBF} = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x}$$

$$\text{where } x = \frac{\text{Mission time}}{\text{MTBF}}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

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6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Combination protector. Combination protector is defined as a protective device which is responsive to motor temperature and current and which protects the motor from overheating due to overload or failure to start by automatically disconnecting the motor from the source of power supply.

6.3.4 Inherent motor protection. A hermetic compressor is considered to have inherent motor protection when it is equipped with a combination protector applied to the motor in such a manner that it protects the motor from overheating due to reduction or loss of cooling medium, overload, failure to start, or phase failure by automatically disconnecting the motor from the source of power supply.

6.4 Abbreviations. In order to save space and to avoid the spelling out of repetitious words and phrases, the following abbreviations have been used in the text of this specification:

BTU/hr.	British thermal units per hour
db	decibels
fpm	feet per minute
g	gravitational unit
Hz	Hertz, cycles per second
kw	kilowatt
MTBF	mean-time-between-failure
NEMA	National Electrical Manufacturers Association
psig	pounds per square inch gage
Ult	ultimate
V	volt

6.5 Identification of changes. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Air Force - 11

Reviewer:
Air Force - 82

Preparing activity:
Air Force - 11

Project No. 4130-F174

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

AFLC–WPAFB–OCT 67 2M

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MIL-C-38392A(USAF)
AMENDMENT 1
9 March 1970

MILITARY SPECIFICATION

COMPRESSORS, HERMETIC, AIR CONDITIONING,
GENERAL SPECIFICATION FOR

This amendment forms a part of Military Specification
MIL-C-38392A(USAF) dated 11 June 1969

Page 1, paragraph 1.2: Delete references to "Class 1", "Type I", "Type II", and all pertinent data. Under Class 2, delete references to "Type II" and "Type III" and all pertinent data.

Page 3, paragraph 3.3.2, line 2: Delete "2,740" and substitute "a minimum of 3,990".

Page 5, paragraph 3.4.1.1, first sentence: Delete "1, 2, and 3" and substitute "2 and 3".

Page 5, paragraph 3.4.2, first sentence: Delete "1, 2, and 3" and substitute "2 and 3".

Page 5, table II: Delete references to "Class 1, types I and II" and all pertinent data. Under Class 2, delete "types II and III" and all pertinent data.

Page 6, paragraph 3.4.4, first sentence: Delete "1, 2, and 3" and substitute "2 and 3".

Page 7, paragraph 3.4.5.6, line 6: Delete "class 1" and connector "MS3102-16S-1P".

Page 8, table III: Delete reference to "class 1" and all pertinent data.

Page 16, paragraph 4.6.16.1: Add "After completion of the tests specified in table VI, the compressor shall be subjected to the tests specified in 4.6.16.3."

Custodian:
Air Force - 11

Reviewers:
Air Force - 82
Navy - MC

Preparing activity:
Air Force - 11

Project No. 4130-F175

FSC 4130

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NOTICE OF CANCELLATION

INCH-POUND

MIL-C-38392A
NOTICE 1
20 October 1994

MILITARY SPECIFICATION

COMPRESSORS, HERMETIC, AIR CONDITIONING, GENERAL SPECIFICATION FOR

MIL-C-38392A, dated 11 June 1969, is hereby cancelled without replacement.

Preparing activity:
Air Force - 82

Project 4130-0368

AMSC N/A

FSC 4130

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MIL-C-38404D
6 Nov 85
SUPERSEDING
MIL-C-38404C
27 Nov 1968

MILITARY SPECIFICATION

COUPLINGS, HOSE, REATTACHABLE SCREW-ON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers screw-on type reattachable couplings for hoses conforming to MIL-H-6615, MIL-H-26521, as specified (see 6.2).

1.2 Classification. The couplings shall be of the following classes and types, as specified (see 6.2):

Class 1	Fuel service
Type I	External NPT thread
Type II	Internal NPSH thread with swivel
Class 2	Demineralized water service
Type I	External NPT thread
Type II	Internal NPSH thread with swivel

* 1.3 Part numbering. Definitive part numbers shall be in accordance with 6.3.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio ALC/MMEDO, Kelly AFB, TX 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4730

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SPECIFICATION

FEDERAL

QQ-P-416	Plating, Cadium (Electrodeposited)
ZZ-H-466	Hose Assembly, Metal (Rubber Covered), Hydrocarbon Conveying.

MILITARY

MIL-H-775	Hose, Hose Assemblies, Rubber, Plastic, Fabric or Metal (Including Tubing); and Fittings, Nozzles and Strainers.
MIL-H-6615	Hose Assemblies, Rubber, Fuel and Water, with Reattachable Couplings, Low Temperature.
MIL-R-6855	Rubber, Synthetic, Sheets, Strips Molded or Extruded Shapes.
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-H-26521	Hose and Hose Assembly, Nonmetallic, Fuel, Collapsible, Low Temperature with Reattachable Couplings.

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

DOD-STD-100	Engineering Drawing Practices.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of.
MIL-STD-810	Environmental Test Methods .
MIL-STD-831	Test Reports, Preparation of.
MIL-STD-1188	Commercial Packaging of Supplies and Equipment.

* (Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein.

NATIONAL BUREAU OF STANDARDS

Handbook H28	Screw-Thread Standards for Federal Services

(Copies may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402).

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AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.1 Alloy and Temper Designation System for Wrought Aluminum.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see sections 4 and 6).

3.2 Components. The coupling shall consist of the following major components:

Item name
Body (see 3.7.1)
Grip (see 3.7.2)
Sleeve (see 3.7.3)

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the corrosion-resistant type or treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service and shall be entirely suitable for the purpose. Materials shall be free from all defects and imperfections that might affect the serviceability of the finished coupling.

3.4.1 Material for class 1 couplings. The material for class 1 couplings shall be as follows:

Body and sleeve	High-grade brass composition, containing not less than 58 percent copper.
Grip	Steel wire, cadmium plated in accordance with QQ-P-416, type II, class 2.

3.4.2 Material for class 2 couplings. The material for class 2 couplings shall be as follows:

Body and sleeve	300 series stainless steel, type B214 or 356 aluminum casting alloy, or aluminum alloys other than 2000, 4000, and 7000 series in accordance with ANSI H35.1. Aluminum alloys shall be anodized in accordance with MIL-A-8625. The color shall be olive drab, color number 24087 of FED-STD-595.

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Grip	Steel wire, cadmium plated in accordance with QQ-P-416, type II, class 2.

3.5 Design and construction. The couplings shall be designed and constructed for use with hoses conforming to MIL-H-6615 or MIL-H-26521. They shall be reattachable, shall incorporate the feature specified herein and as shown on figure 1, and shall require no cements, sealants, or adhesives in their installation, other than to seal the cut ends of the hose.

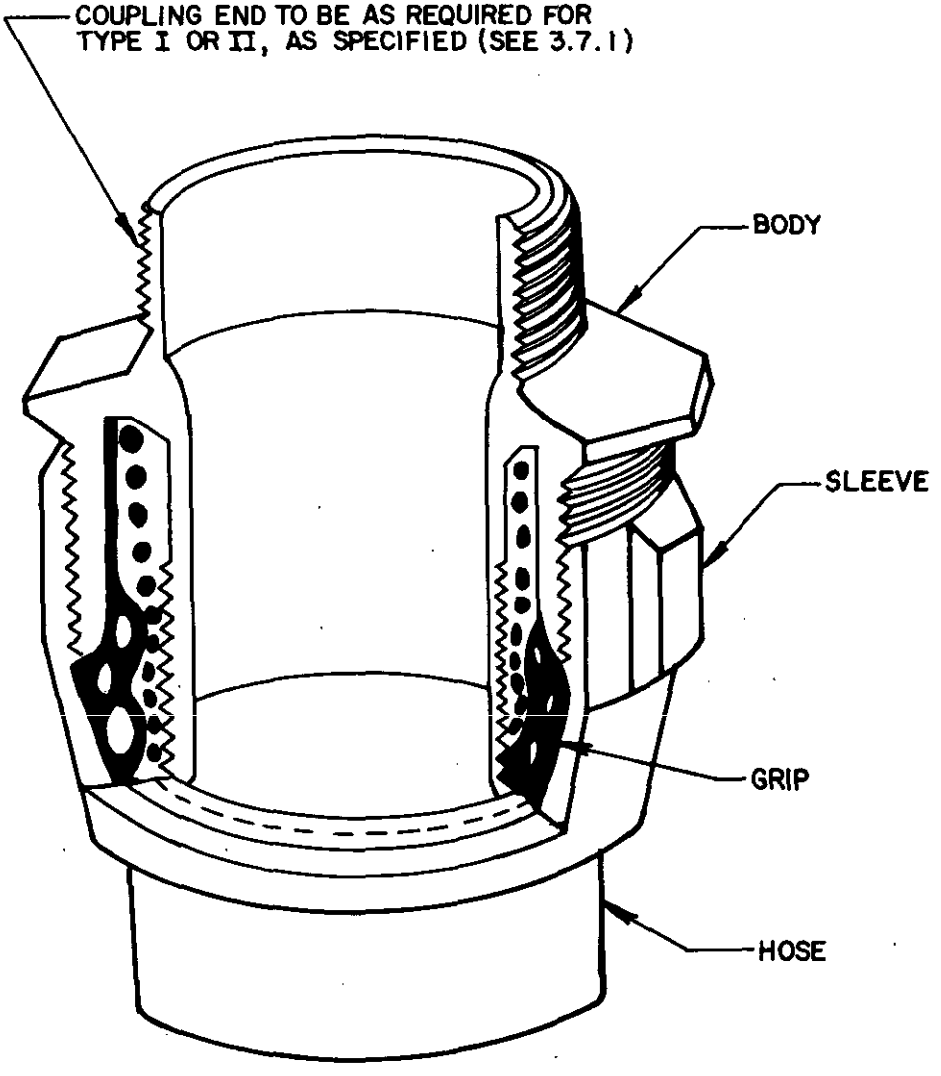


FIGURE 1. Section showing typical coupling assembly construction

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3.5.1 Thread gages. All threads shall be in accordance with Handbook H28. If the male tailpiece threads and female sleeve threads are in accordance with the Handbook H28 special threads (NS or UNS), thread gages shall be provided by the manufacturer. Data specifying the accuracy of the thread gages shall be made available to the procuring activity. (See 3.7.1.1, 3.7.1.2, 3.7.1.3, and 3.7.3).

3.6 Performance.

3.6.1 Environmental conditions. The couplings shall exhibit no corrosion after having been exposed to the following environmental conditions:

a. Relative humidity up to 100 percent, including conditions wherein condensation takes place in the form of water and frost.

b. Salt atmosphere as encountered near bodies of salt water and in transportation on shipboard.

3.6.2 Hydrostatic pressure rating. The internal hydrostatic pressure rating shall be as specified in Table I. The size of the coupling shall be as specified by the procuring activity (see 6.2).

* 3.6.3 Hose retention. The couplings shall be installed on the hose and shall not be detached from the hose when subjected to a repeated longitudinal load. After the repeated load test, couplings shall not blow off or leak when test pressure specified in Table I.

TABLE I. Pressure rating.

Size	Size code	Working pressure (psi)	Test pressure (psi)
1.25	A	400	600
1.50	B	400	600
2	C	380	570
2.50	D	360	540
3	E	320	480
4	F	280	420

3.7 Detail of components.

3.7.1 Body. The body of the coupling shall have a configuration as shown on Figure 2. The body shall have a coupling end for attachment to mating assemblies and a tailpiece end for attachment of the applicable hose. The central hexagon or octogon shall have a larger dimension across the flats than the largest diameter across corners of the coupling sleeve.

3.7.1.1 Type I coupling end. The type I coupling end shall have external American Standard taper pipe threads (NPT). The end surface of the coupling end shall be a smooth finish with rounded edges, square with the thread axis, to provide a good sealing surface when joined to a type II coupling.

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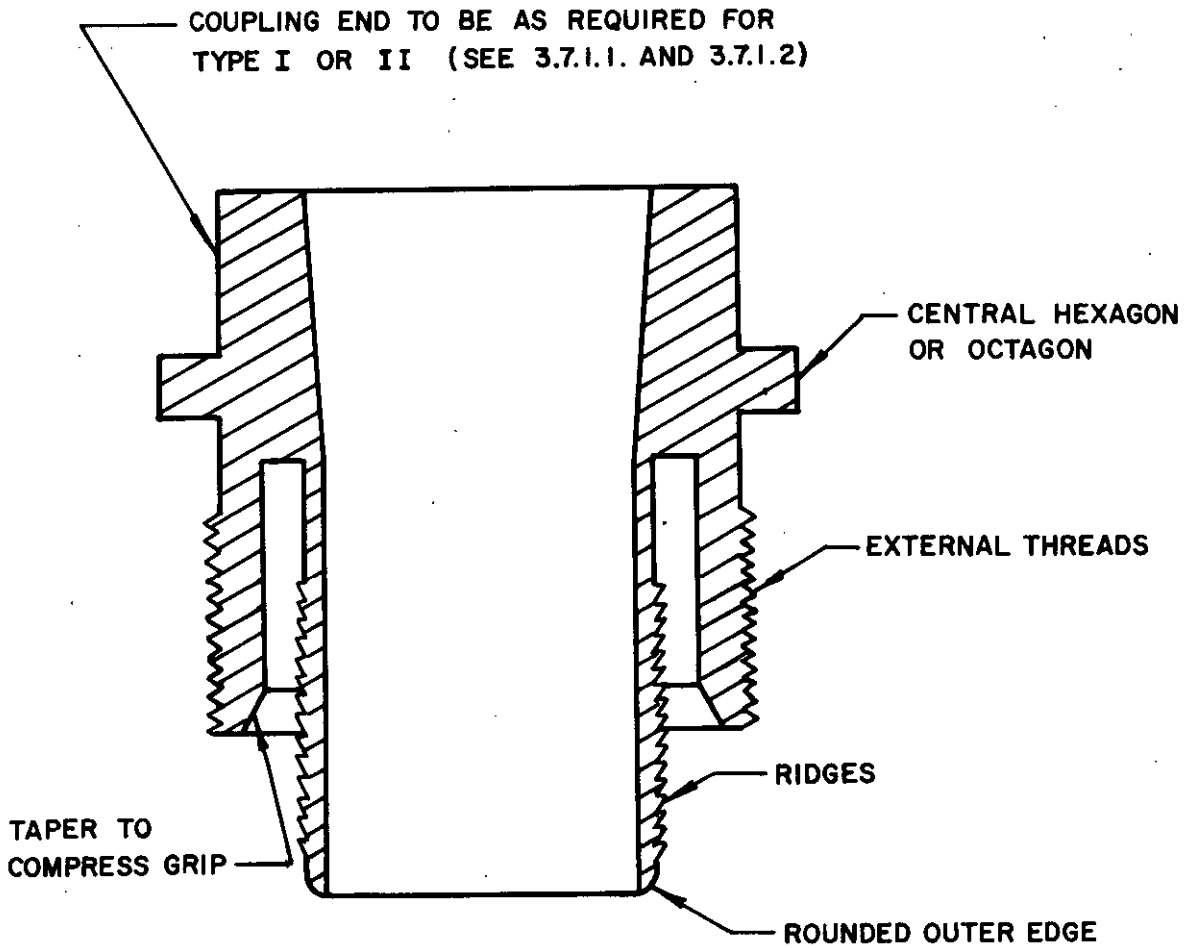


FIGURE 2. Typical cross-section of coupling body

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3.7.1.2 Type II coupling end. The type II coupling end shall have internal American National hose coupling threads (NPSH) which shall swivel on the remainder of the body. The 2 1/2-, 3-, and 4 inch nominal size type II couplings shall conform to the 8 threads per inch sizes of ZZ-H-466. The body shall provide a smooth sealing surface for a gasket. Each body shall be furnished with a gasket of synthetic rubber conforming to MIL-R-6855 class 1. The swivel end shall turn freely by hand, without the use of tools or mechanical assistance, until slightly before a leaktight seal is effected against the gasket.

3.7.1.3 Tailpiece end. The tailpiece end shall consist of two concentric walls spaced to accept hose in accordance with MIL-H-6615 or MIL-H-26521, of the appropriate size. The outer wall of the tailpiece end shall have external threads to engage the sleeve, and the inside end shall be tapered to compress the grip. The external threads shall be straight and no coarser than 10 threads per inch. The inner wall shall have ridges to assure a tight grip on the hose and shall have a rounded outer edge to prevent cutting of the hose during installation or upon sharp bending.

3.7.2 Grip. The grip shall be of material as specified in 3.4.1 and 3.4.2. The wire shall be left-hand wound into a coil of approximately three loops, and may be round or flat with rounded edges. The end of the grip shall be ground to prevent damaging the hose or injuring using personnel during installation or removal of the coupling.

3.7.3 Sleeve. The sleeve shall have a configuration incorporating the features shown on Figure 3 and as specified herein. Four flat surfaces (spaced 90 degrees apart) shall be provided for attaching a wrench. The interior of the sleeve shall have internal threads for connection with those provided on the body, shall be tapered to compress the grip, and shall have the edge rounded to prevent cutting of the hose upon sharp bending. When fully threaded to the body, the sleeve shall extend beyond the body tailpiece.

3.8 Interchangeability. All functionally and dimensionally interchangeable part shall have the same manufacturer's part number. The drawing number requirements of DOD-STD-100 shall govern changes in the manufacturer's part numbers.

* 3.9 Marking. The sleeve specified in 3.7.3 shall be marked (stamped) with the applicable hose size and manufacturer's code in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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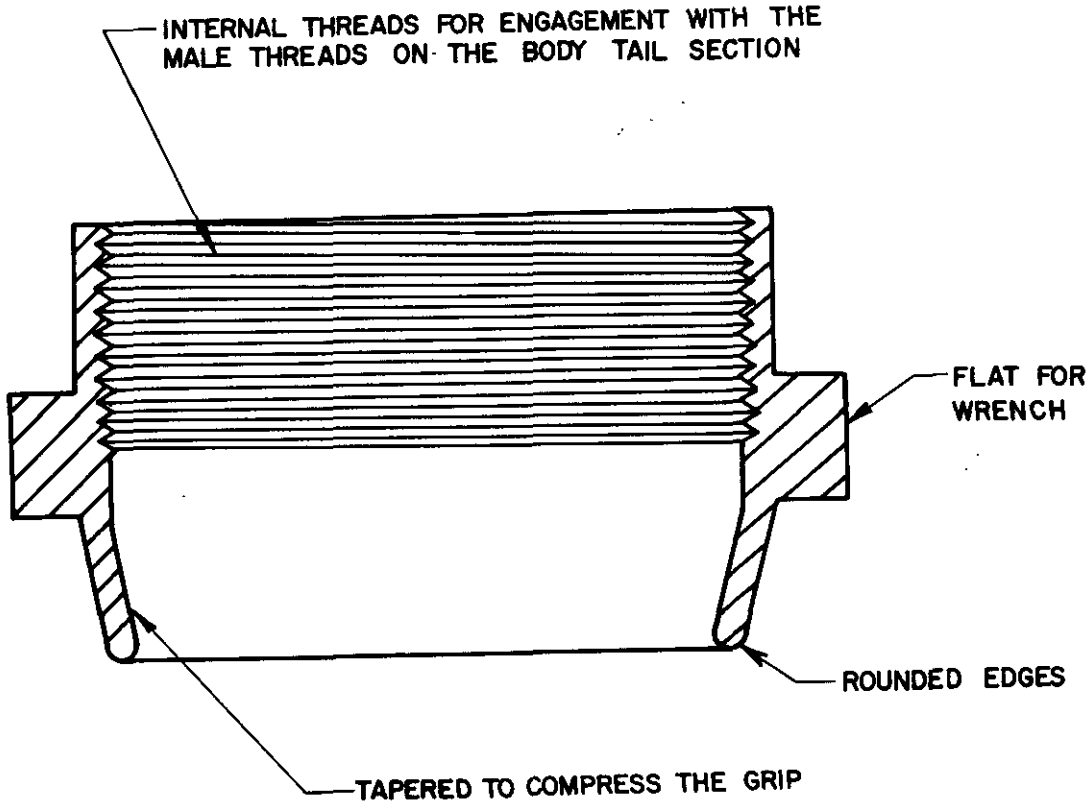


FIGURE 3. Cross-section of coupling sleeve

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4.2 Classification of inspection. The examination and testing of the couplings shall be classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection (see 6.2).

4.3.1 Test samples. One coupling of each class and type being procured for each applicable hose specification shall be subjected to the tests specified in 4.3.3.

4.3.2 Test report. Upon completion of the first article tests, three completed copies of a test report in accordance with MIL-STD-831 shall be supplied to the procuring activity.

4.3.3 First article inspection. The first article inspection shall consist of all the tests described under 4.5.

4.4 Quality conformance inspection. The quality conformance inspection shall consist of the individual tests.

4.4.1 Individual tests. Each coupling shall be subjected to the examination specified in 4.5.1. If the body is cast, each coupling shall be subjected to the test specified in 4.5.2.

4.5 Inspection methods.

4.5.1 Examination of product. The coupling shall be inspected to determine compliance with the requirements specified herein. The following shall be applicable to the first article item only and shall be included in the test report:

a. Weight of component.

b. Dimensions of components.

c. Certification that the materials used complied with the requirements specified herein.

4.5.2 Pressure test. The coupling body shall be subjected to the applicable internal hydrostatic test pressure specified in Table I for not less than 1 minute. Evidence of leakage shall be cause for rejection.

4.5.3 Swivel test. When applicable, the internally threaded end connection of the body shall demonstrate the ability to swivel by hand relative to the tailpiece which is attached to the hose. After this demonstration, the coupling assembly shall be subjected to the test specified in 4.5.4.

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* 4.5.4 Hose retention test. The coupling assembly shall be attached to a six foot long section of hose conforming to MIL-H-6615. The hose assembly shall be hung vertically by any convenient means and be used to lift a dead weight of 450 pounds at least one inch off the floor and then lowered. This cycle shall be repeated 25 times. The assembly shall then be subjected to the test pressure specified in Table I for a minimum of one minute. Any evidence of leakage, pull or blow off of the coupling shall be cause for rejection. The above shall be repeated with a six foot long section of hose conforming to MIL-H-26521.

4.5.5 Environmental tests. The coupling shall be tested in accordance with 4.5.5.1 through 4.5.5.3.

4.5.5.1 Humidity. The coupling shall be subjected to humidity in accordance with Procedure I, Method 507, of MIL-STD-810.

4.5.5.2 Salt spray. The coupling shall be subjected to salt spray in accordance with Procedure I, Method 509, of MIL-STD-810.

4.5.5.3 Verification with substantiating data that materials used are not adversely affected by humidity or corrosion may be provided in lieu of the humidity and salt spray testing.

4.6 Packaging, packing, and marking. Preparation for delivery shall be inspected for conformance to section 5.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with applicable levels of MIL-H-775 or MIL-STD-1188, as specified (see 6.2).

5.2 Marking. In addition to any special marking required by the contract or order (see 6.2), interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The couplings covered by this specification are intended for use on hose conforming to MIL-H-6615 or MIL-H-26521 in ground fueling of aircraft with hydrocarbon fuels (use class 1) or demineralized water (use class 2).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class, type and size (see 1.2, 3.5, 3.6.2, 3.7.1, and figure 1).

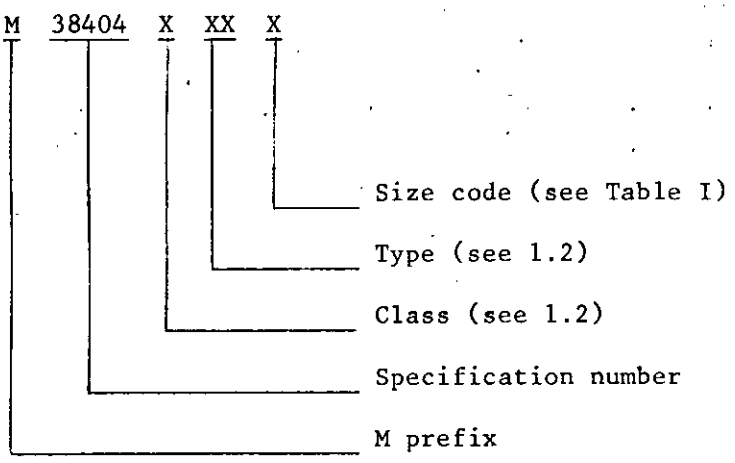
c. Location and conditions for first article inspection (see 4.3).

d. Applicable levels of preservation, packaging, and packing required (see section 5).

e. Special shipment marking (see 5.2).

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6.3 Part number. Definitive part number shall be in accordance with the following:



Example:

M 38404-2-II-C

Class 2, type II, 2 inch I.D.

* 6.4 Changes from previous issue. The margins of this document are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractor's are cautioned to evaluate the requirements of this document based on the entire content, irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99
Army - ME

Review activity:
Army - AV
DLA - CS

Preparing activity:
Air Force - 82

(Project 4730-0566)

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INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-38404D	Couplings, Hose, Reattachable Screw-On

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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MILITARY SPECIFICATIONS

CLOTH, PLAIN WEAVE, NYLON

1. SCOPE

1.1 This specification covers the requirements for one type cloth, nylon, mesh, color black (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods
PPP-P-1133	Packaging and Packing of Synthetic Fiber and Silk Fabrics

STANDARDS

Federal

FED. STD. No. 4 Glossary of Fabric Imperfections

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

FEDERAL TRADE COMMISSION

Textile Fiber Products Identification Act, Effective March 3, 1960

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

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3. REQUIREMENTS

3.1 Standard sample. The cloth shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 Material

3.2.1 Yarn. The yarn for the warp and filling shall be monofilament nylon having a diameter of 0.008 plus or minus 0.001 inch.

*3.3 Color. The color of the cloth shall be lusterless black, shade 221.

3.3.1 Colorfastness. The finished cloth shall show fastness to perspiration and laundering equal to or better than the standard sample. When a standard sample is not established or not designated as applicable to colorfastness, the finished cloth shall show good fastness to perspiration and laundering when tested as specified in 4.4.

*3.3.2 Matching. The color of the fabric shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard shade under incandescent lamplight at 2800° Kelvin.

*3.4 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.

*TABLE I. Physical requirements

Weight per square yard		Yarns per inch Minimum		Breaking strength Minimum		Weave
Ounces				Warp	Filling	
Min	Max	Warp	Filling	Pounds		----
4.8	5.2	50	35	200	175	Plain

3.4.1 Width. The finished minimum width of the cloth, inclusive of selvage, shall be as specified (see 6.2).

3.5 Finish. The cloth shall be heat treated at 350° ±5°F to achieve a stabilized cloth in accordance with the requirements specified in 3.6.

3.6 Shrinkage. The cloth shall not shrink more than 2.0 percent in the warp direction nor more than 1.5 percent in the filling direction when tested as specified in 4.4.

*3.7 Length. Unless otherwise specified (see 6.2), the cloth shall be in one continuous piece not less than 50 yards and shall be put up in rolls as specified in PPP-P-1133.

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3.8 Fiber identification. Each roll shall be ticketed, labeled, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act, Effective March 3, 1960.

3.9 Workmanship. The finished cloth shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the point level specified.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Examination of the end item. Examination shall be in accordance with the provisions of 4.2.1 through 4.2.4.

*4.2.1 Yard-by-yard examination. Thirty yards from each roll in the sample shall be examined on one side. The number of rolls from which the sample yardage shall be extracted shall be in accordance with the following:

Lot Size (Yards)	Sample Size (Rolls)
3200 or less	10
3201 up to and including 10,000	15
10,001 on over	25

An approximately equal number of yards shall be examined in each roll of the sample. The lot shall be unacceptable if the points per hundred yards exceed 30. Point computation shall be as follows:

$$\frac{\text{Total points scored in sample} \times 3600}{\text{Sample size (yards)} \times \text{contracted width of material (inches)}} = \text{Points per square yard.}$$

No more than one roll shall be taken from any shipping container unless the number of shipping containers in the lot is less than the required number of rolls, in which case all shipping containers shall be present in the sample. All defects, as defined in section III of Fed. Std. No. 4, that are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.1.1. No linear yard (increment of 1 yard on the measuring device of the inspection machine) shall be penalized more than 4 points.

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4.2.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects up to 3 inches in any dimension	- one point
For defects exceeding 3 inches, but not exceeding 6 inches in any dimension	- two points
For defects exceeding 6 inches, but not exceeding 9 inches in any dimension	- three points
For defects exceeding 9 inches in any dimension	- four points

Note: The following defects, when present shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth

Width less than specified

Objectionable odor

Poor dye penetration, mottled, streaky, or cloudy.

4.2.2 Examination for length

4.2.2.1 Individual rolls. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or two yards below the length marked on the ticket shall be considered a defect with regard to length. The lot shall be unacceptable if two or more rolls in the sample are defective in respect to length.

4.2.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls in the sample is less than the total of the lengths marked on the tickets.

4.2.3 Examination for compliance with Textile Fiber Products Identification Act. During the yard-by-yard examination, each piece in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each piece not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

4.2.4 Examination for shade. During the yard-by-yard examination, each roll shall be examined for shade. Two or more rolls in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

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4.3 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of PPP-P-1133 to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation.

4.4 Tests. The method of testing specified in CCC-T-191, wherever applicable and as listed in table II shall be followed. The physical values specified in section 3 apply to the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 3 yards full width of the finished cloth. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The sample size (number of sample units) and lot size expressed shall be in accordance with the following:

LOT SIZE (in yards)	SAMPLE SIZE
800 or less	2
801 to 22,000	3
22,001 and up	5

TABLE II. Test methods

Characteristic	Requirement reference	Test method
Yarn		
Monofilament	3.2.1	Visual 1/
Diameter	3.2.1	3
Fiber identification	3.2.1	3
Weave	Table I	Visual 1/
Yarns per inch	Table I	5050
Weight	Table I	5041
Breaking strength	Table I	5100
Colorfastness to:		
Perspiration	3.3.1	5680
Laundering	3.3.1	5614
Shrinkage	3.6	5556 2/

1/ One determination shall be made on each sample unit and the result reported as "pass" or "fail."

2/ Cotton laundering procedure in test method 5556 shall be used.

3/ The supplier shall submit a certificate of compliance for this characteristic. The certificate shall be accompanied by actual test, inspection, or other verifiable quality data.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

*5.1.1 Levels A and C. The cloth, put up as specified, shall be packaged in accordance with the applicable requirements of PPP-P-1133.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2 and 6.2.1).

*6.2.1 Levels A, B, and C. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1133.

*5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1133.

6. NOTES

6.1 Intended use. The cloth covered by this specification is intended for use as a stiffener for the brim and crown of the Hat and Insect Net HGU-18/P (MIL-H-38471).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Width of cloth required (see 3.4.1)

c. Minimum length, when other than specified (see 3.7)

d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or purchase orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-B-1133, may be exercised in the preparation for delivery of the cloth.

6.3 Standard sample. For access to standard sample address the procuring activity issuing the invitation for bids.

6.4 The cloth covered by this specification was previously procured under U. S. Army LP/P.DES 13-63.

Custodian:
Air Force - 11

Preparing activity:
Air Force - 11

Review Activities
Air Force - 82, 85

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *IF "YES", IN WHAT WAY?*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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NOTICE
OF CANCELLATION

INCH-POUND

MIL-C-38469A(USAF)
NOTICE 1
27 SEPTEMBER 1991

MILITARY SPECIFICATION

CLOTH, PLAIN WEAVE, NYLON

MIL-C-3849A(USAF), dated 2 February 1967, is hereby cancelled without replacement.

Preparing Activity:
AIR FORCE - 71

(Project 8305-F400)

AMSC N/A

FSC 8305

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